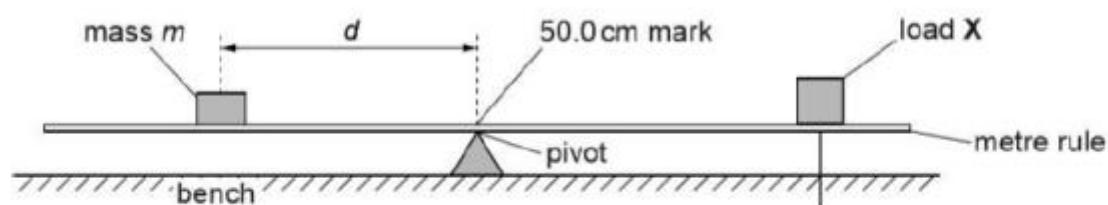
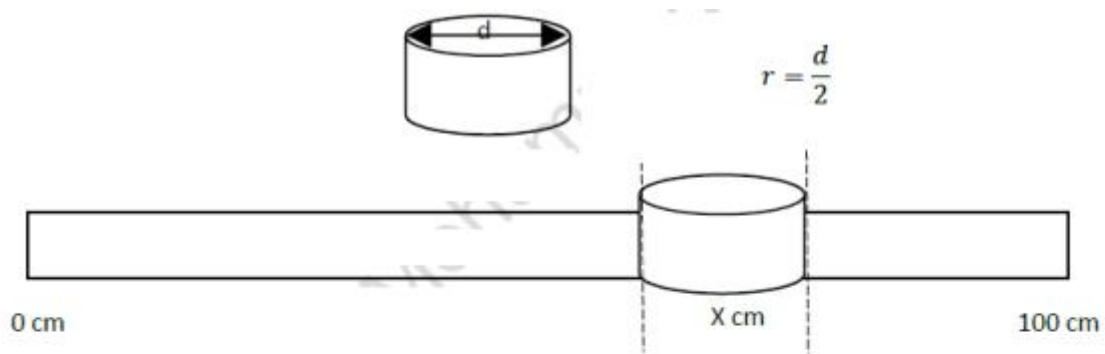


Physics- Alternate to Practical

Balancing Experiment



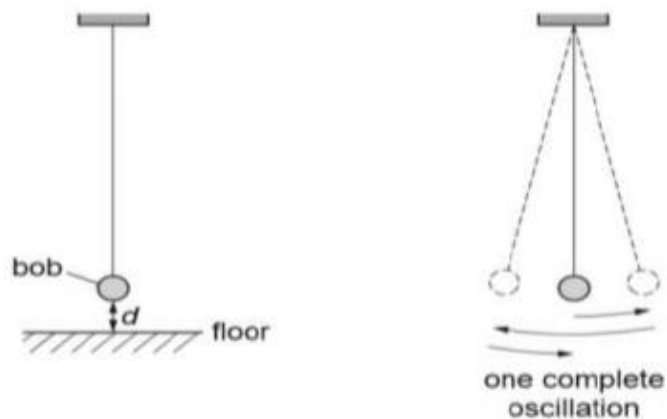
Difficulty	Way to overcome
Balancing the ruler above the pivot	repeat several times until it balances
The mass slips over the ruler:	Stick the mass with the ruler
The center of mass of the cylinder may not be above the correct mark (X)	- Measure the diameter of the cylinder. - Adjust the cylinder above the correct mark (X) so that one side of cylinder is at mark $X-r$ and the other at mark $X+r$



Precautions	Repeat and take average
	Look perpendicular to the scale of the ruler while taking the reading to avoid parallax error
	Range of masses

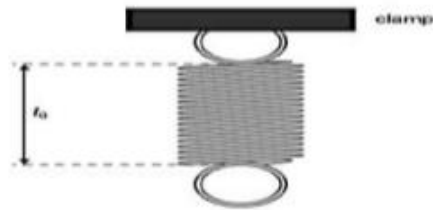
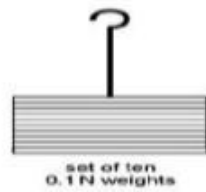
Mass of the ruler	From 100 g to 200 g
Position of center of mass	The point at which the meter rule is balanced above the pivot.

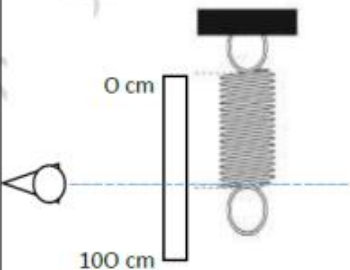
Pendulum Experiment



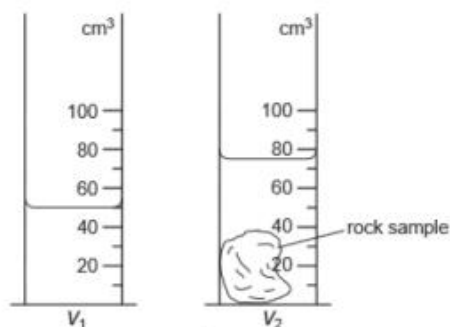
Precaution	- No air current in the place. - Take the time for number of swings then divide the time measured by the number of swings - Repeat and take average. - Look perpendicular to the ruler while measuring the height
Conditions kept constant to fair comparison	- Same stopwatch. - Release the bob from the same height. - Same number of swings. - Same shape of bob. - Same size of bob.

Spring Experiment



To make a fair comparison between springs of different materials	- Same diameter. - Same Thickness of spring. - Same range of loads. - Same Length of wire. - Same thickness of wire. - Same number of coils - Same coil spacing
Precautions	- Look perpendicular to the scale of ruler while taking readings. - Waite until the spring stops vibrating.
Position of meter rule	

Displacement Method Experiment



Inaccuracy	1. Parallax error. 2. Student didn't look from the meniscus of the liquid. 3. Splashes during immersing the rock. 4. Air bubbles may found in the rock. 5. String used may be thick. 6. Cylinder mayn't be sensitive.
Precautions	- Shake the rock. - Thin string used. - Look perpendicular to the scale of the cylinder. - Look from the meniscus. - Put the rock gently.

Inaccuracy	- Thick string. - Turns aren't tangent. - Thick marking. - Spacing between turns.
Precautions	- Thin string used. - Tangent turns without spaces. - Thin marks. - Look perpendicular to the scale of ruler. - Take more number of turns.

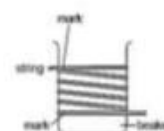
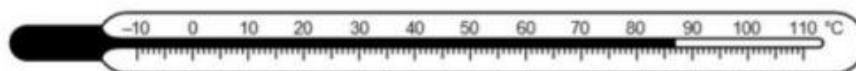


Fig. 1.4

The string is held against a metre rule as shown in Fig. 1.5

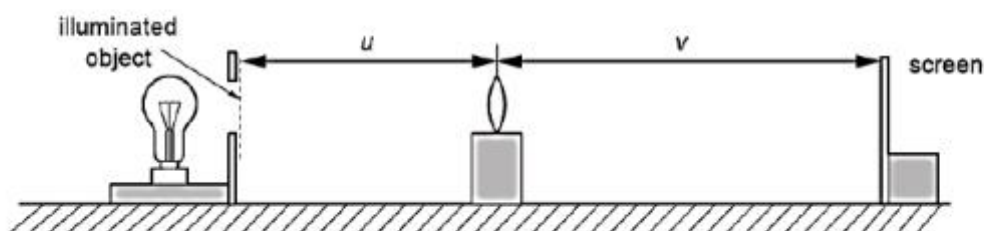


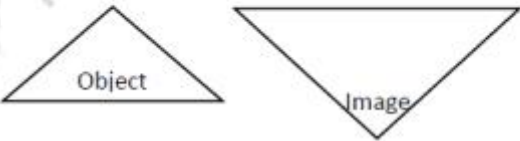
Thermal Experiment



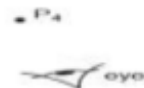
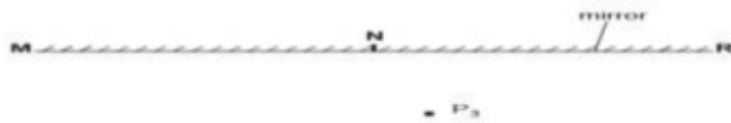
Precautions	Look perpendicular to the scale of the thermometer while taking the reading to avoid parallax error
Conditions that should be kept constant when this experiment is repeated.	- Room temperature. - Starting temperature. - Same volume of water. - Same beaker - Same thermometer. - Air current.
Inaccuracy of this type of experiments	Heat lost to surrounds
Overcoming heat lost	- Lag container with insulator. - Cover the container with lid - Initial temperature near to room temperature.

Optical Experiment I



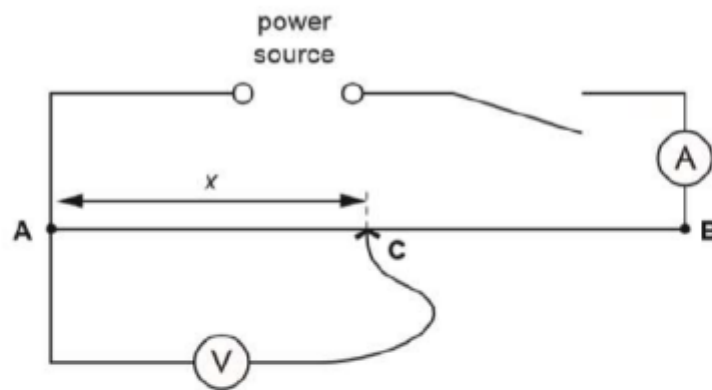
Precautions	- Dark room - Object and lens same height from bench - Mark on lens holder to show position of lens center. - Take more readings. - Repeat and take average. - Look perpendicular while taking readings.
Image formed	- Inverted. - Magnified if: <i>Image distance > object distance.</i>  - Smaller: <i>Object distance > Image distance.</i>

Optics Experiment I I



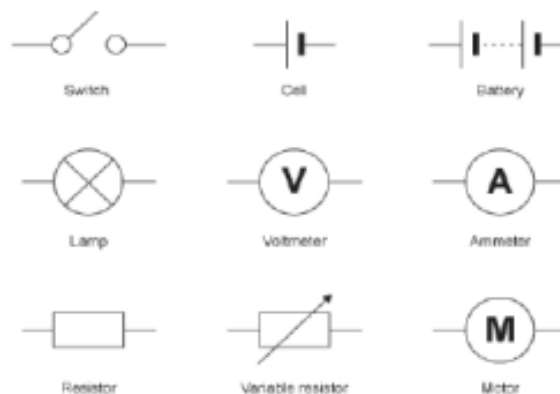
Suggest possible causes of this inaccuracy	- Thickness of lines. - Thickness of protractor - Thickness of pins. - Pin holes - Thickness of mirror. - Glass in front of mirror causes double refraction.
Precaution	- Place the pins as far apart as possible. - Draw the lines so that they are as thin as possible - Look perpendicular while taking readings to avoid parallax error. - Look from the base of the pins: a. No concern about pins being vertical. b. Base of the pin lie on the rays. c. Base is always perpendicular to the plane.

Electricity Experiment



Inaccuracy	- Heating effect of the current. - Battery used up. - Bad connection of sliding contact C
Precautions to overcome heat in the wire	- Use a battery of lower electromotive force. - Switch on and off between readings.
Precautions in General	- Look perpendicular while taking readings. - Tap on the voltmeter and ammeter to check that pointer free to move. - Take several readings then take overall average.

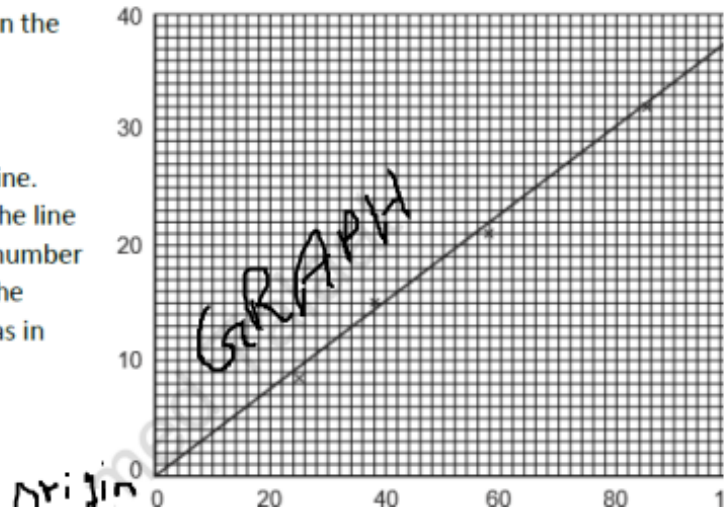
Electricity Symbols



Graphs

Graph Conditions:

- 1- Correct labels on the axes.
- 2- Suitable scale (you don't have to start with origin).
- 3- Graph should be at least more than the half of the work sheet.
- 4- Plot points correctly (x) or 
- 5- Draw best fit line:
 - Most of the points placed on the line.
 - If all the points aren't located on the line draw a symmetric line (make the number of points above the line equal to the number of points below the line) as in figure.
- 6- Thin continuous line.



Gradient calculations:

- 1- Choose two point from the line drawn so that the distance between them at least equal to half the distance of line
- 2- Show the taken point with dashed lines or triangle method.
- 3- Use the equation:

$$G = \frac{Y_2 - Y_1}{X_2 - X_1}$$

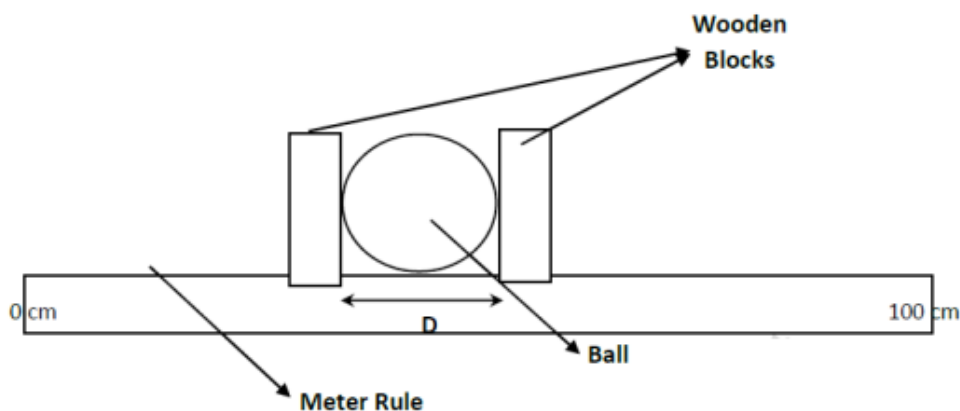
The direct proportional between the two variables if:

- A. The graph:
 - 1- Graph shows straight-line with positive gradient.
 - 2- The line passing through the origin point.
- B. The ratio between the two variable is constant within the experimental accuracy.

The inverse proportional between the two variables if:

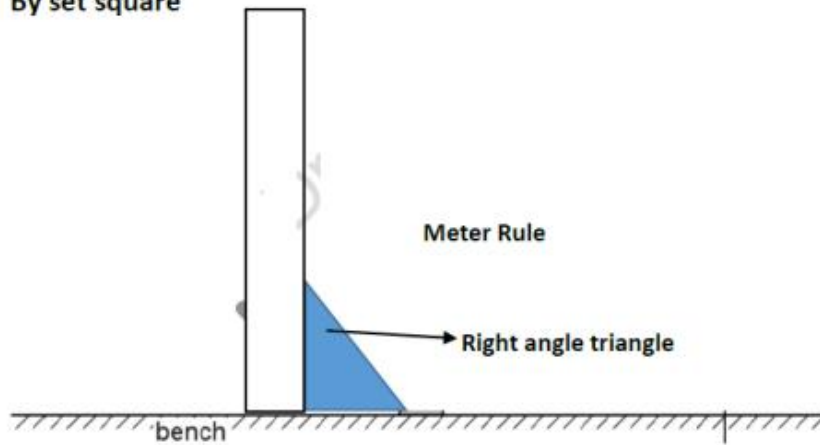
The multiplication between the two variable is constant within the experimental accuracy.

Measuring the diameter of a ball



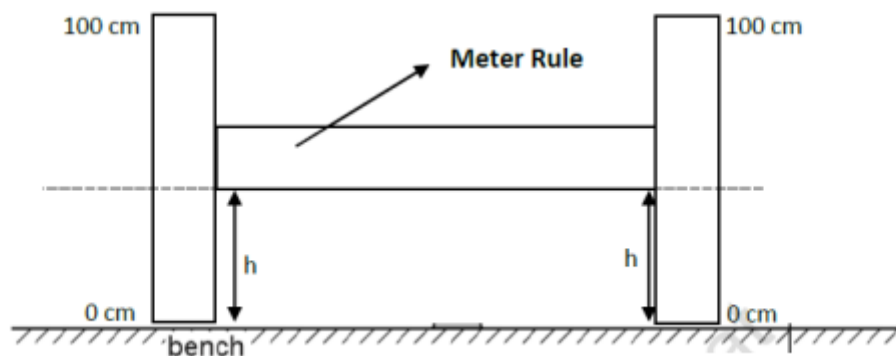
How to check that the ruler is vertical to the bench?

By set square



How to check that the ruler is Horizontal to the bench?

Measure the height of the meter rule above the bench from both sides



How to compare between two results to know the experimental limit?

- 1- Divide the greater number by the smaller one.
- 2- Multiply the decimal of the result by 100 to the percentage error.

If the percentage is below 10%:

The results are equal within the experimental accuracy.

If the percentage is beyond 10%:

The results are not equal within the experimental accuracy.

Example

An experiment is performed to know the focal length of a lens by two methods and the results are:

$$f_1 = 15 \text{ cm}$$

$$f_2 = 14.7 \text{ cm}$$

Apply the rule:

Step one: $\frac{15 \text{ cm}}{14.7 \text{ cm}} = 1.02$

Step two: $\% = 0.02 \times 100 = 2\%$

So the results are equal within experimental accuracy as the difference between the results is 2% which is below the experimental limit.

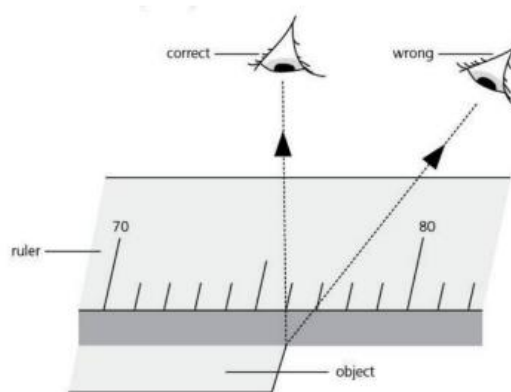
1. Skills regarding handling the experiment

A. How to measure the length?

a) Measuring the length of an object:

i. The ruler (up to 30 cm)

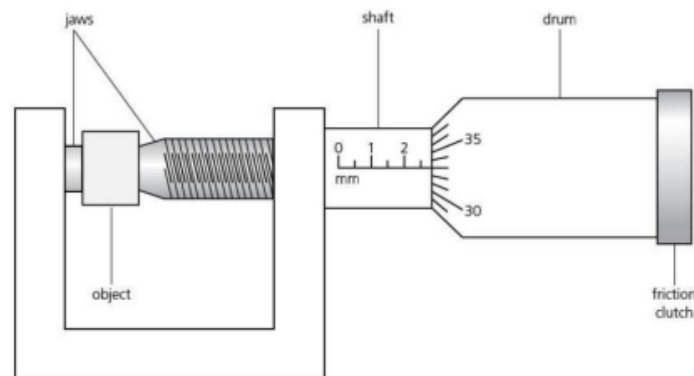
- Accuracy: $0.1 \text{ cm} = 1 \text{ mm}$
- The ruler should be placed as close to the object as possible.
- The eye must be directly above the mark on the scale or the thickness of the ruler causes parallax error.



- When measuring extensions (of springs, for example), it is best to record the actual scale readings for the stretched and relaxed lengths, and then work out the extension afterwards.

II. The micrometre screw gauge (up to 2 cm)

- Accuracy: $0.01 \text{ mm} = 0.001 \text{ cm}$
- Can be used to measure small objects in the millimetre range.
- One revolution on the drum (50 divisions) = one division on the shaft (0.5mm)
i.e. one drum division = $0.5/50 = 0.01 \text{ mm}$ (accuracy).
- The total reading = shaft scale reading + drum scale reading $\times$ (accuracy)
- A friction clutch ensures that the jaws always exert the same force on an object and over-tightening does not occur.
- Before making measurement, check to ensure that the reading is zero when the jaws are closed, otherwise a zero error must be allowed for where a reading is taken.



The object length = 2.5mm on the shaft scale + 33 division on the drum scale =
 $2.5 \text{ mm} + (33 \times 0.01 \text{ mm}) = 2.5 \text{ mm} + 0.33 \text{ mm} = 2.83 \text{ mm}$

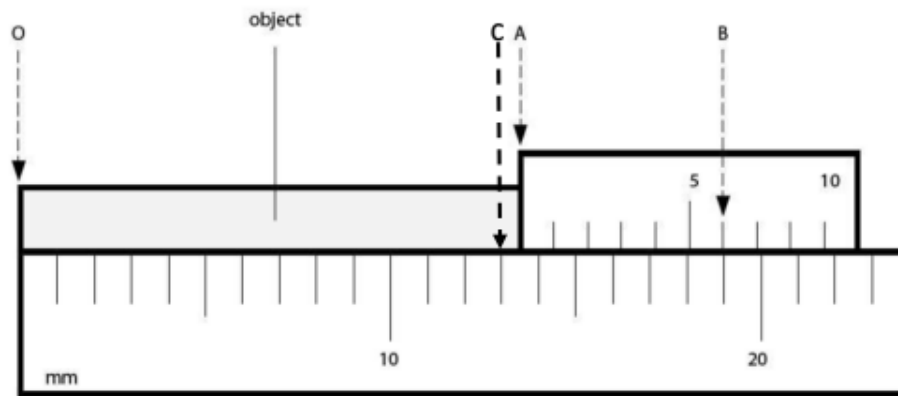
III. Vernier scale (up to 15 cm):



- Accuracy: $0.1 \text{ mm} = 0.01 \text{ cm}$
- Has two scales, the mm scale and Vernier scale (9.0 mm divided into 10 divisions 10.0 equal divisions). One Vernier division = 0.9 mm.
- The correct reading = the reading on the mm scale at coincidence – the number of marks on Vernier scale at coincidence $\times 0.9 \text{ mm}$ OR
The correct reading = The reading on mm scale at the end of the object + the number of marks on Vernier scale $\times 0.1 \text{ mm}$.

The object length = OB – $(6 \times 0.9 \text{ mm}) = 19 \text{ mm} - 5.4 \text{ mm} = 13.6 \text{ mm}$ OR

The object length = OC + $(6 \times 0.1 \text{ mm}) = 13 \text{ mm} + 0.6 \text{ mm} = 13.6 \text{ mm}$



IV. The meter rule (up to 1 m)

Accuracy: 1 mm = 0.1 cm

V. The measuring tape (distances more than 1 m)

Accuracy: 1 cm

Notes:

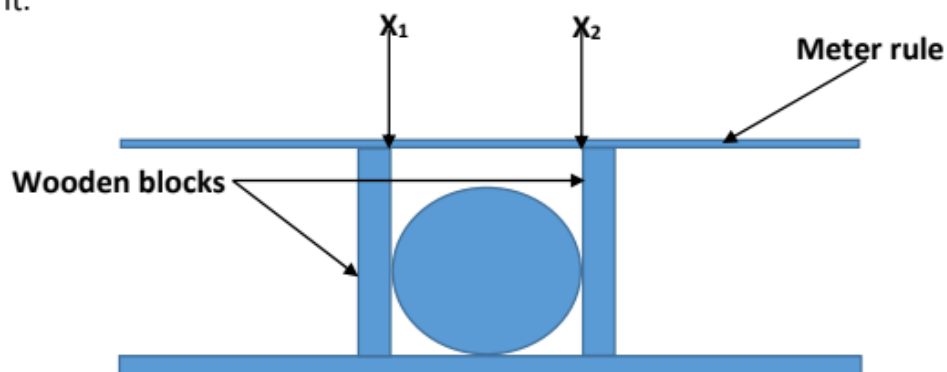
- each reading should be repeated several times to get an acceptable average.
- Some readings should be excluded if their values are very far from the other readings (abnormal readings).

b) Measuring thickness of a thin object:

Multiples can be measured and then divided to find an average value. For example, to obtain the average thickness of one page of a book, measure the thickness of 20 pages and divide your result by 20.

c) Measuring the diameter of a cylinder or a sphere:

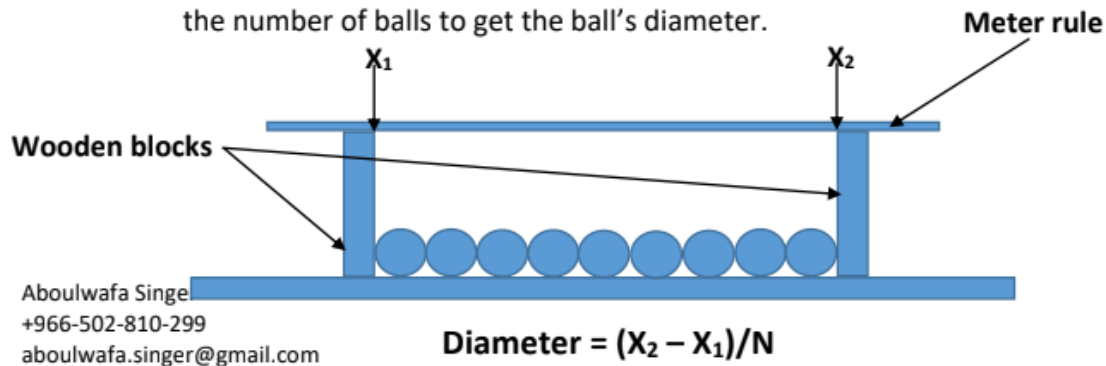
To measure the diameter of a sphere or a cylinder, place it between two vertical blocks and take the difference between the two readings at the edges of it.



$$\text{Diameter} = X_2 - X_1$$

d) Measuring Diameter of ball bearings:

To measure the diameter of ball bearings collect them together in a row between two vertical blocks, measure the total length and then divide by the number of balls to get the ball's diameter.

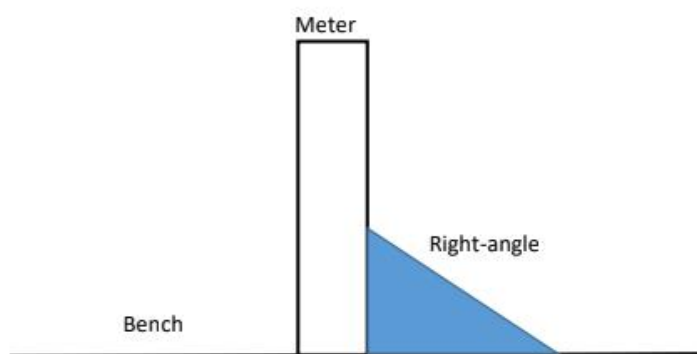


5



e) How to check that the ruler is vertical to the bench?

- By using set square method: put the set square (right-angle triangle) so that one side of the angle is parallel to the bench, while the other side of the right angle is parallel to the meter rule.



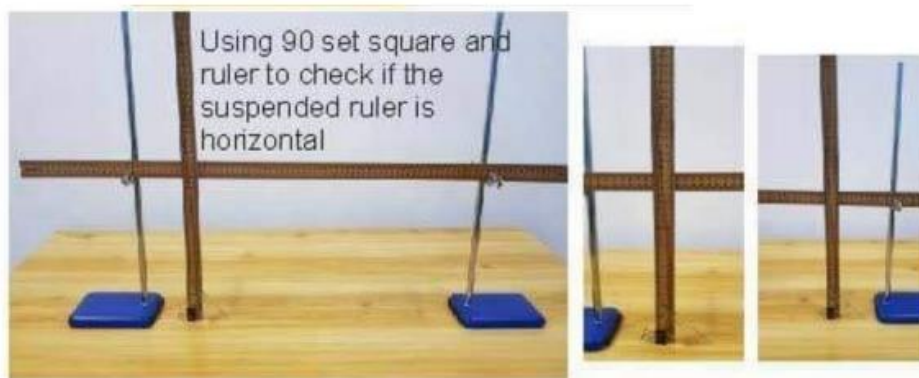
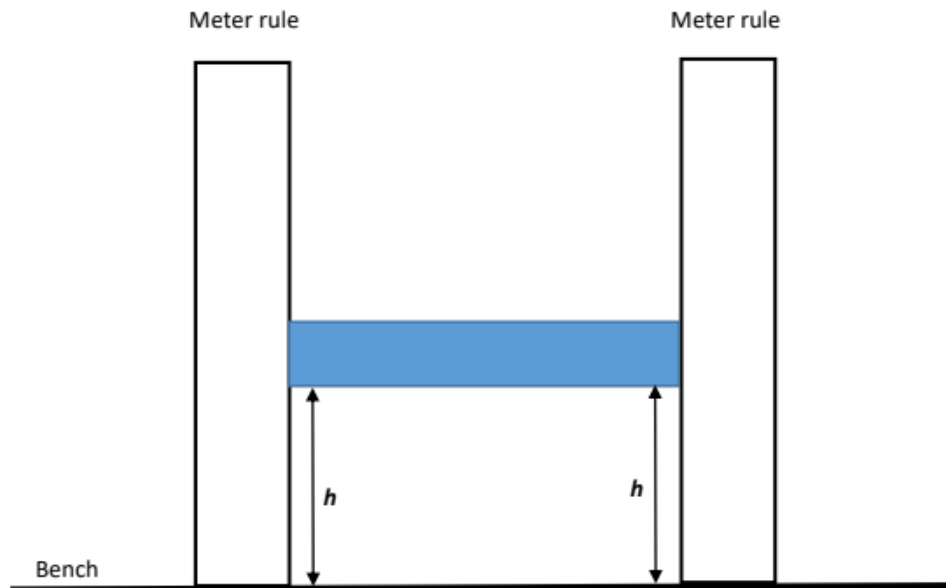
- By protractor

f) How to check that the ruler is horizontal to the bench?

- By using the spirit level



- By measuring the height of the meter rule above the bench (h) from both sides (must be equal).



B. How to measure a circumference of a cylindrical-shaped object (cup, test tube...)?

To measure a circumference of a cylindrical-shaped object, wind a string 10 times around it then unwind it, measure the length of the string and divide by 10. The string should be thin, tangent and without spacing between turns.



- **Reasons for inaccuracy:**

- Using thick string
- The marks are thick
- Leaving space between turns
- Winding turns at angles
- Stretching of the string

- **Precautions:**

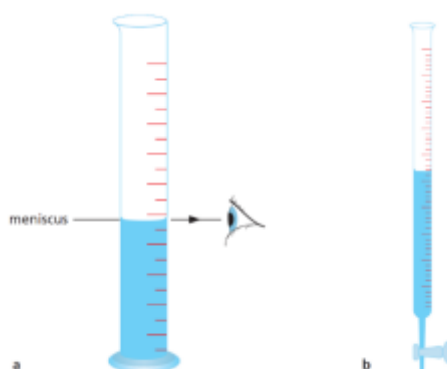
- Use thin string
- Thin marks
- Tangent turns without spacing
- Take more number of turns
- Make sure that the string isn't stretched

Note: Dividers could be used if there is a difficulty using the measuring tool directly or when the distance is not easy-to-access. The divider is kept fixed and then measured by the rule.

C. How to measure the volume?

1) Volume of liquids

- The volume of relatively large volumes is measured using measuring cylinder (a). Small volumes are measured using a pipette or a burette (b).
- The unit of volume is ml. $1\text{ml} = 1\text{cm}^3$, $1\text{l} = 1\text{dm}^3$ and $1\text{m}^3 = 10^6\text{m}^3$.
- The following precautions should be taken into consideration:
 - The measuring cylinder must be vertical by setting on a horizontal bench.
 - The eyes should be perpendicular to the scale to avoid parallax
 - The reading should be taken at the *bottom* of the meniscus of the liquid surface unless for *mercury*, it should be taken at the *top* of the meniscus.



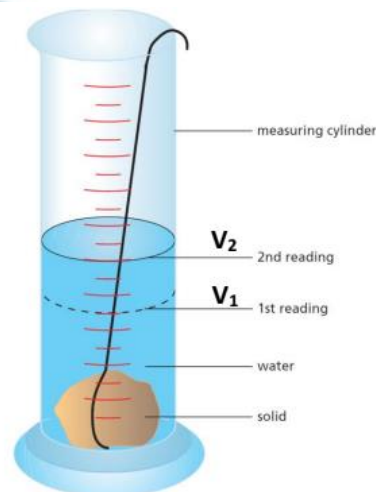
2) Volume of regular solids:

- Volume of a rectangular block = Length $\times$ width $\times$ Height
- Volume of a sphere of radius $R = \frac{4}{3}\pi R^3$
- Volume of a cylinder of radius R and height $H = \pi R^2 H$

3) Volume of irregular solids:

➤ By using *measuring cylinder* (Accuracy 1cm^3):

- Fill the measuring cylinder with some water with a volume V_1 .
- Insert the solid gently in the cylinder and make sure it is totally immersed.
- Measure the volume of the solid and the water V_2 .
- The volume of the solid is $V_2 - V_1$.

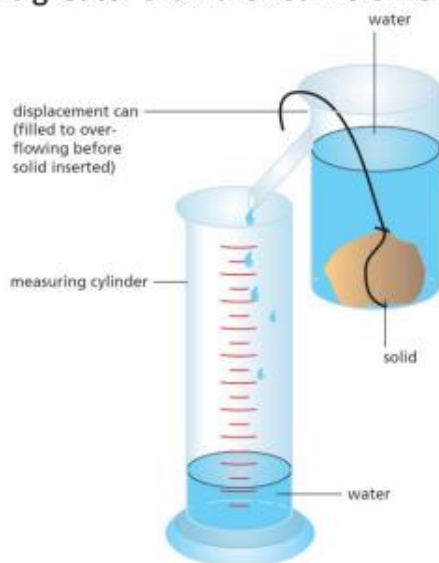


➤ By using *displacement can and measuring cylinder*:

When the solid is large to be inserted in the cylinder.

- Fill the displacement can to its full capacity.
- Insert the solid gently in the displacement can and collect the volume of water that floods from the can in an empty measuring cylinder.
- Take the volume reading in the measuring cylinder which is the volume of the solid.

- **Note:** in both methods, if the solid traps air inside it so that some volume of air cannot be occupied by the liquid, the measured volume is greater than the real volume of the solid.



D. How to use measure the mass?

1. Mass can be measured by a digital top-pan balance, which gives a direct reading of mass placed on the pan.
2. The unit of mass is $\text{kg} = 1000 \text{ g}$
3. The mass of a very light object like a pin is determined by finding the mass of large number of these objects and then divided by the number of object to get the average mass of a single object.
4. Note:
 - The balance should be clean, horizontal and reads zero before applying a mass on it.
 - A digital top-pan balance is **accurate** to the size of the smallest mass which can be measured on the scale setting you are using, probably 1 g or 0.1 g.



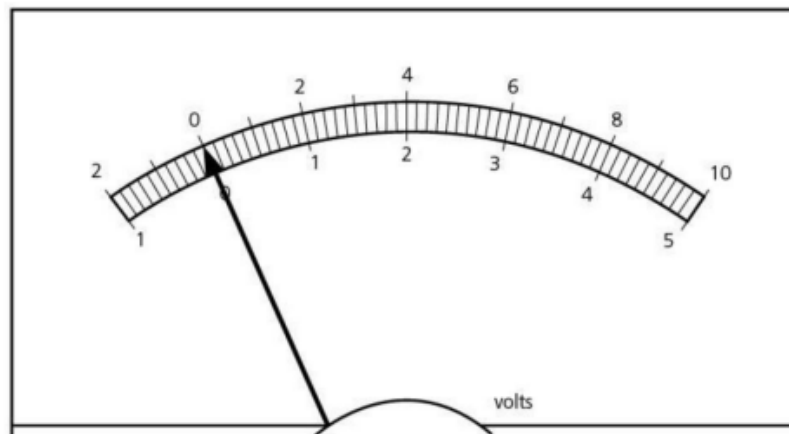
E. How to measure the time?

- Long times (hours) are measured using a clock or a watch.
- Shorter times (up to one hour) are measured using a stopwatch. It should be reset to zero before pressing start: otherwise a correction should be performed.
- Very short times (up to s sec) are measured using electronic /digital timer



F. How to read results on Ammeter and Voltmeter?

- An *ammeter* measures electric current
- The unit of current is ampere (A)
- An ammeter should be placed *in series* with the device in which the current is to be measured.
- A *voltmeter* measures potential difference (p.d.).
- The unit of potential difference is the volt (V).
- A voltmeter should be placed *in parallel* with the device across which the p.d. is to be measured.
- There are two types of Ammeters/Voltmeters, *analogue* and *digital*.

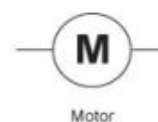
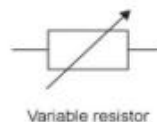
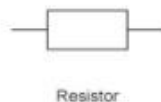
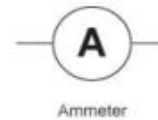
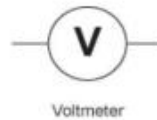
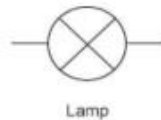
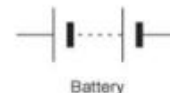
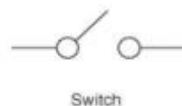


- If the analogue meter is used, the accuracy = 1 division value.
- The division value = difference between two successive numbers on the scale/number of divisions between them.
- For the upper scale in figure the division value= $(2-0)/10 = 0.2 \text{ V}$ = the accuracy.
- For the lower scale in figure the division value= $(1-0)/10 = 0.1 \text{ V}$ = the accuracy.
- As with rulers, the eye should be immediately above the pointer when taking a reading to avoid introducing parallax errors; if there is a mirror behind the pointer, the needle and its image should coincide when you take a reading.

- Check that the meter reads zero when there is no current; adjust the screw at the base of the pointer until it reads zero.
- Digital meters allow different ranges to be selected and the display gives measurement in whatever units have been chosen.
- The reading will be **accurate to the last figure on the display**, so for small currents and voltages it will be more accurate to use the mA or mV setting.
- **Tips for using meters in electrical circuits:**
 - Construct circuits with the power switched off or battery disconnected and the meter last.
 - Check that meters are connected with the correct polarity and are set to their **lower sensitivity** initially.
 - Set the power supply output to zero before you switch it on.



G. What are the electricity symbols?



2. Skills regarding handling the data

A. Calculation skills:

a) Significant Figures and rounding

Note: Always show answers to 2 or 3 significant figures. Double check all calculations at the end of the test and check that you have given the correct unit.

- When doing calculations, your answer should have the same number of significant figures as the measurements used in the calculation. For example, if your calculator gives an answer of 1.23578, this should be given as 1.2 if the measurements on which you based your calculations have two significant figures and 1.24 if your measurements have three significant figures.

➤ **Example: if you measure lengths in mm, then:**

- When you use a rule, no SF is allowed to the right of the decimal point as the accuracy is only 1mm. 15.7 mm should be written as 16.0 mm.
- When you use a Vernier scale, only 1 SF is allowed to the right point as the accuracy is only 0.1 mm. 15.73 mm should be written as 15.7mm.
- When you use a micrometre, only 2 SFs are allowed to the right of the decimal point as the accuracy is only 0.01 mm. 15.73 mm should be written as 15.73 mm and 15.77mm is rounded to 15.80 mm.
- If values of different numbers of SFs are used to calculate a quantity, quote your answer to the smallest number of SFs.
- If a number is expressed in standard notations, the number of significant figures is the number of digits before the power of 10; for example, 6.24×10^2 has three significant figures.
- In deciding the least SF, you look at the following figure; if that is less than 5. You round down (1.23 becomes 1.2) but if it is 5 or above, you round up (1.235 becomes 1.24).

b) **Calculating the average according to SFs:**

- Sum the values for a quantity you have measured and divide the sum by the number of values to obtain the average.
- For example if you measure the length of a pendulum as 81.5 cm and 81.6 cm, then:

$$\text{The average value} = \frac{(81.5+81.6)}{2} \text{ cm}$$

$$\begin{aligned}
 &= \frac{(163.1)}{2} \text{ cm} \\
 &= 81.55 \text{ cm} \\
 &= 81.6 \text{ cm}
 \end{aligned}$$

The value has been given to 3 SFs because that was the accuracy of the individual measurements on which the calculation was based.

c) Experimental accuracy limit (below & beyond)

- Subtract the smaller number from the greater number.
- Divide the result by the greater number
- Multiply the result by 100 to obtain the percentage error.
- If the percentage is below 10%, then the results are **equal within the experimental accuracy**.
- If the percentage is beyond 10%, Then the results are **not equal within the experimental accuracy**.
- Example: An experiment is performed to determine the focal length of a lens by two different methods.

$$f_1 = 16.0 \text{ cm}$$

$$f_2 = 15.5 \text{ cm}$$

$$\% = \left(\frac{\text{the greater value} - \text{the smaller value}}{\text{the greater value}} \right) \times 100$$

$$\% = \left(\frac{16.0 - 15.5}{16.0} \right) \times 100 = 3\%$$

- The results are equal within the experimental accuracy as the percentage (3%) is less than 10% (below the experimental accuracy limit).

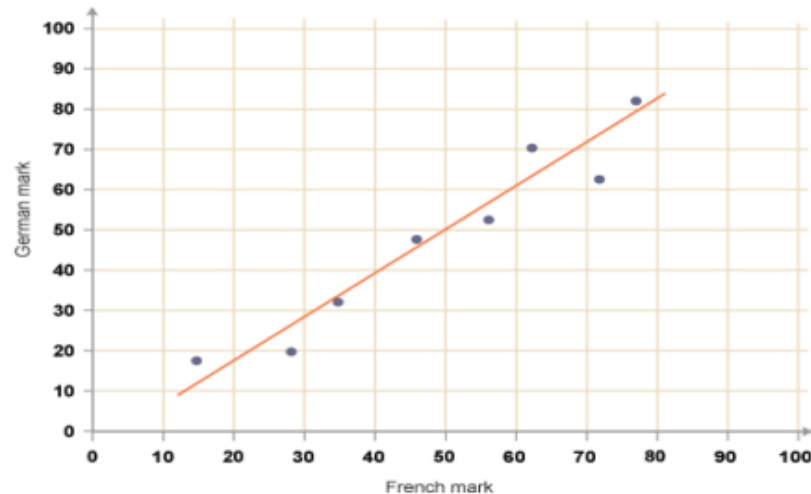
B. Graphing skills

a) How to best represent and plot the data?

- You will need about **6 points taken over as large a range as possible** to plot a graph.
- Choose scales that make it easy to plot the points and use **as much of the graph paper as possible (at least half of X axis and half of Y axis unless more)** but don't use an awkward scale to force the graph to take up all the grid. If your range of values is for example 89 to 170, you don't need to start at zero, start at 80.
- Make sure you **label each axis of graph with the name and unit** of the quantity being plotted e.g. Resistance / Ω (Don't write Resistance (Ω), use the forward slash "/").

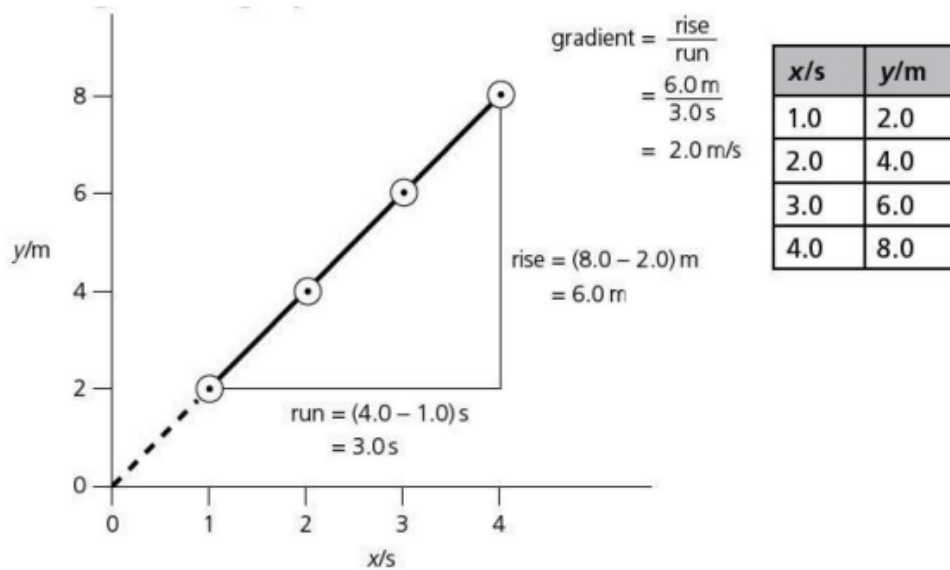
- Plot all points to within $\frac{1}{2}$ small square (examiner checks this very carefully).
- Mark the data points clearly with a ***dot in a circle or a cross with a sharp pencil.***
- Join up your points with a ***single, continuous, sharp and smooth line or curve.***

b) How to make a line fitting?



- In practice, points plotted on a graph from actual measurements may not lie exactly on a straight line due to experimental error.
- The best straight line is then drawn through them such that they are equally distributed about it; this automatically averages the results.
- Do not force the line through the origin unless the graph should go through the origin (for example this is fine in the case of someone starting a stopwatch at a start line if you are plotting time against displacement).
- If possible, repeat any anomalous measurement/calculation to check that they have been recorded and/or calculated properly or try to identify the reason for the anomaly.

c) How to calculate the gradient/slope (triangle method)?



- The slope or gradient of a straight-line graph can be determined by the triangle method.
- Use as long length of line as possible to determine the gradient from the ratio of the vertical (rise) to the horizontal (run) of the triangle chosen.
- It is best practice to draw the triangle in your exam paper with a thin line.

➤

$$\text{Gradient} = \frac{\text{rise}}{\text{run}}$$

$$= \frac{6}{3} \text{ m/s}$$

$$= 2 \text{ m/s}$$

- Show your working out – put numbers into $(y_2 - y_1) / (x_2 - x_1)$

d) How to describe the relation represented by the graph?

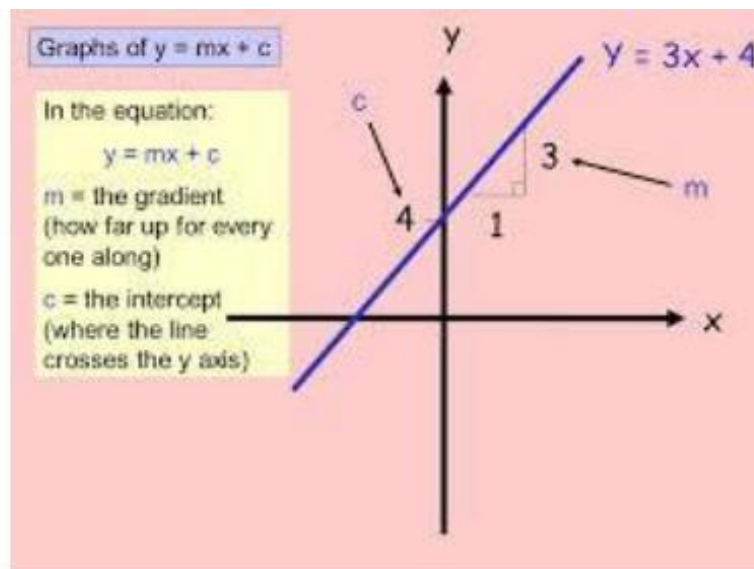
i. Linear:

- The straight line equation is:

$$Y = mx + c$$

where, m is the slope/gradient and c is constant. The constant c represents the intersect point with the Y axis (at $X = 0$).

- X and Y are called to have a linear dependence relation.



II. Direct proportionality:

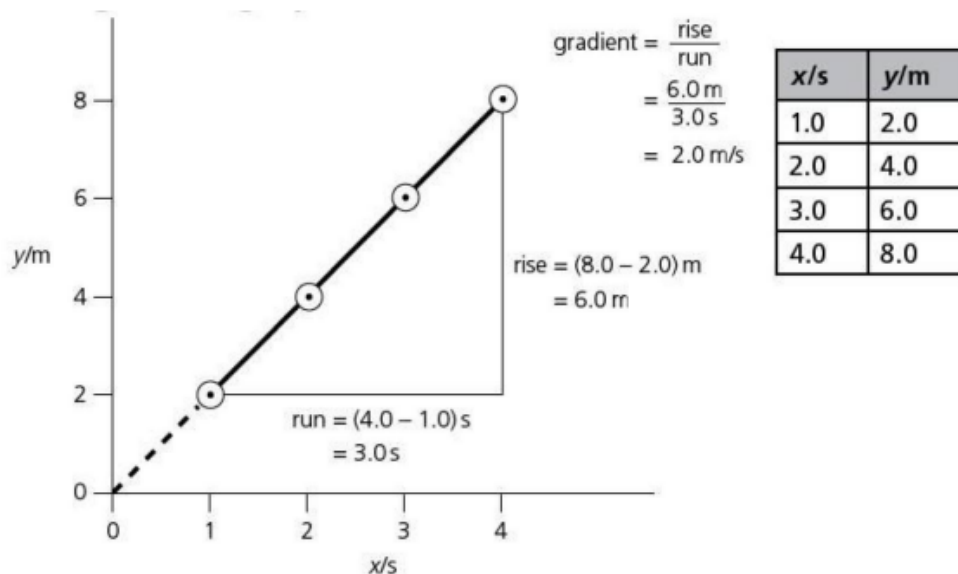
- If the constant in the linear equation becomes zero as a special case, then the linear equation becomes:

$$Y = mX$$

In this case, the line intersects at $X = 0$ and $Y = 0$ (the origin) and the relation is said to be ***DIRECT***.

- Each ***DIRECT*** equation is ***LINEAR*** but not each ***LINEAR*** equation is ***DIRECT***.
- The main sign of the ***DIRECT*** equation is that:

$$\frac{Y}{X} = m \text{ is always constant all over the graph.}$$



- For this graph, $\frac{Y}{X} = 2$ for all the points on the graph.

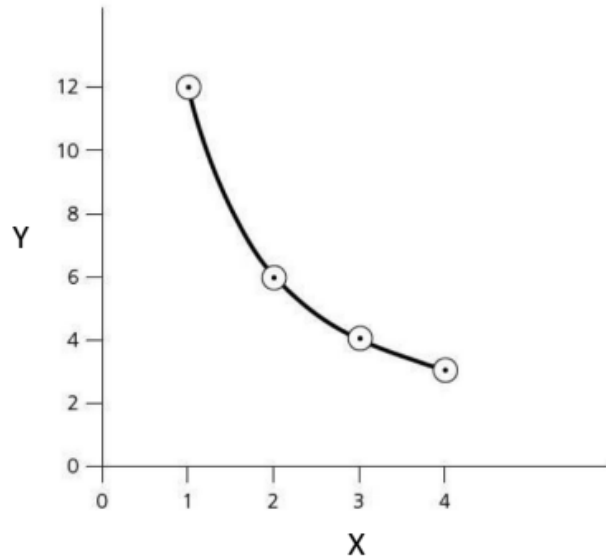
$$(Y \propto X)$$

III. Inverse proportionality:

- If the relation between X and Y is:

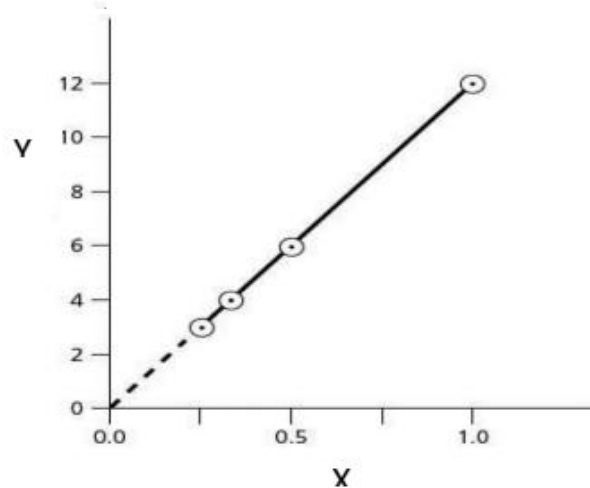
$$Y = \frac{m}{X}, \text{ then } YX = m \text{ is always constant.}$$

- The continuous line in this case is a curve.



x	Y	1/x
1.0	12	1.00
2.0	6	0.50
3.0	4	0.33
4.0	3	0.25

- For the graph above, $YX = 12$ for all the X and Y values. It represents an *INVERSE* proportionality relationship.
- If, however, $1/X$ is plotted against Y (or X is plotted against $1/Y$) then a straight line through the origin is obtained.



- In this case $1/X$ is proportional to Y and there is an inverse proportionality between X and Y :

$$\frac{1}{X} \propto Y \quad \text{OR} \quad \frac{1}{Y} \propto X$$

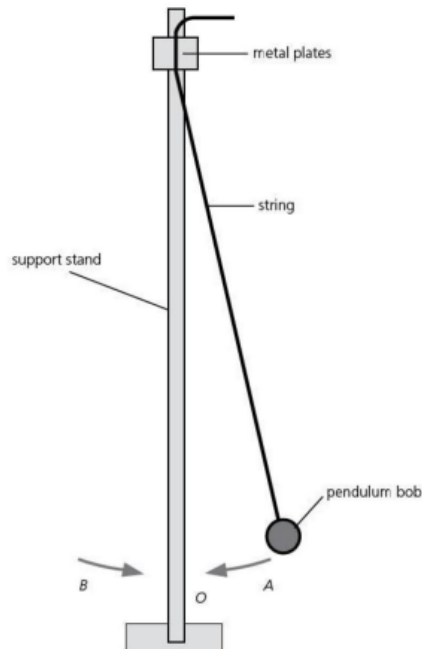
e) Table skills:

- Numerical values should be given to the number of significant figures appropriate to the measuring device.
- Column headings in tables of readings must be headed with the quantity and unit as in these examples: l/A , or t/s , or y/m .
- Any **calculated quantity** should have the **least number of SFs** of the quantities that were involved in the calculation (or at most one more).
- If an **average** is calculated, it should be written to the least number of **decimal places** of the averaged numbers.

Experiments:

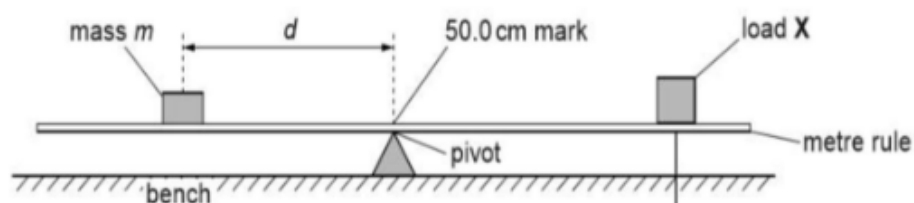
1. General Physics:

1.1 Simple pendulum

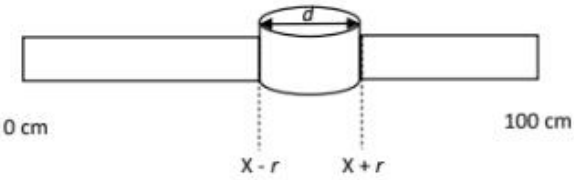
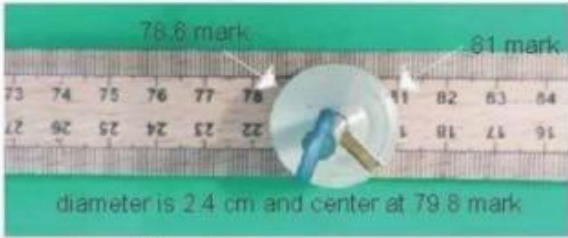


Aim	To investigate the effect of length and mass on the period of a simple pendulum.
Apparatus	Support stand-String-2 metal pendulum bobs of different mass-metre rule-scales.
Procedure	1. Measure the length of the pendulum from the point of support to the centre of the bob; repeat measurements and calculate the average. 2. Measure the time taken for the bob to make ten small angle oscillations; Repeat the measurement and calculate the average time. 3. Work out the period, T, of the pendulum –this is the time needed for one oscillation. 4. Measure the mass of the pendulum bob. 5. Repeat steps 1 to 4 using a longer pendulum length. 6. Repeat steps 2 to 4 using a heavier pendulum bob and the same pendulum length as was used in step 5.
Precautions to improve accuracy or to give more reliable results	▪ No air currents in the place. ▪ Take the time of number of swings then divide the time measured by the number of swings to get T. ▪ Repeat and take the average. ▪ Look perpendicular to ruler while measuring the height. ▪ Use the set-square as a horizontal reference. ▪ Metre rule should be close to pendulum. ▪ Measure the length from the centre of the mass of the bob
Conditions kept constant to fair comparison or to repeat the experiment	▪ Same stopwatch. ▪ Release the bob from the same height (same amplitude). ▪ Same number of swings. ▪ Same shape of bob. ▪ Same size of bob.
Common questions	▪ Explain why measuring time of 20 swings (oscillations) rather than 1 swing is more accurate ➤ To reduce the effect of reaction time ➤ To reduce error ➤ Time of one swing is too small to be measured ▪ Why measuring the time of 200 oscillations isn't suitable? ➤ Pendulum may stop ➤ Student may lose counting ▪ Suggest a practical reason why the result obtained from experiment is different? ➤ Student didn't react quickly when the pendulum started. ➤ Difficulty to measure the length from the centre of bob.
	▪ Suggest improvement to the experiment ➤ Repeat and take average ➤ Use different lengths ➤ Use fiducial mark ➤ Draw a graph for the results to find the anomalous results.

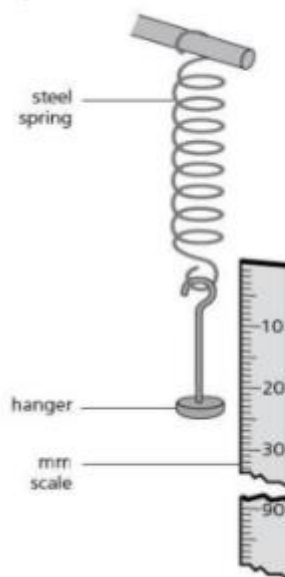
1.2 Balancing a beam:



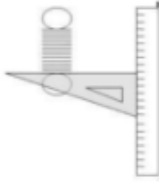
Aim	To measure the moment about a pivot and to show that there is no net moment on a body in equilibrium.
Theory	The moment of a force is a measure of its turning effect and is given by: $\text{Moment of a force} = F \times d$ Where F is the turning force acting on a body and d is the perpendicular distance of the line of action of the force from fulcrum. For a body in equilibrium, the law of moments states that the sum of the anticlockwise moments about any point equals the sum of clockwise moments about the same point.
Apparatus	As shown in figure
Procedure	Set up the apparatus as shown in the figure,
Difficulties and how to overcome them	Balancing the ruler above the pivot: - Repeat several times until it balances The mass slips over the ruler: - Stick the mass with the ruler The centre of mass of the cylinder may not above the correct mark (X): - Measure the diameter of the cylinder d. - Calculate the radius $r = d/2$ - Adjust the cylinder above the correct mark (X) so that one side of the cylinder at mark $(X-r)$ and the other at mark $(X+r)$.

	$r = d/2$  0 cm 100 cm $X - r$ $X + r$  78.6 mark 81 mark diameter is 2.4 cm and center at 79.8 mark
Precautions to improve accuracy or to give more reliable results	▪ Repeat and take the average ▪ Look perpendicular at the scale of the ruler while taking the readings to avoid parallax error. ▪ The range of masses
Common questions	What is the mass of the ruler? ▪ From 100 to 200 g Where is the position of the centre of the mass? ▪ It is the point at which the meter rule balanced above the pivot

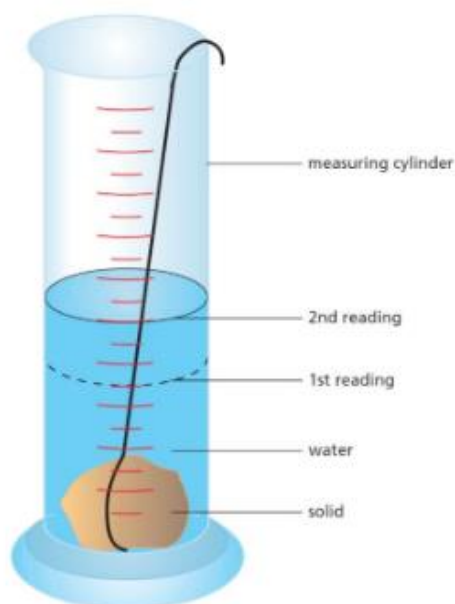
1.3 Spring Experiment (Hook's law):



Aim	To investigate Hook's law for a spring.
Theory	Hook's law states that: $F = kx$ Where F is the stretching force, x is the extension of the spring and k is the spring constant. The law should hold if the spring is not permanently stretched
Apparatus	-retort stand (clamp stand)-spring-hanger with 100 g weights-ruler-adhesive/sticky tape.
Procedure	Set up the apparatus as shown in the figure. 1. Fix the ruler vertically next to the spring so that it can be used as a scale. 2. Record the position of the bottom of the unweighted spring on the scale, l_s, and repeat your measurement. 3. Measure the length of the hanger, l_0. 4. Hang an unweighted (100 g) hanger on the spring and record the scale position of the bottom of the hanger 5. Add a 100 g mass to the hanger and again record the scale position of the bottom of the hanger. 6. Repeat step 5 with 200g, 300g and 500 g masses on the hanger. 7. Plot a graph of the stretching force along the x-axis against the extension along the y-axis. 8. Calculate the gradient of the graph.
Difficulties and how to overcome them	▪ The clamp retort stand might topple. To overcome: use small loads. ▪ Spring might overstretch/spring too weak To overcome: use loads that don't overstretch spring
Possible causes of inaccuracy	▪ Spring extension is not uniform with load ▪ Spring exceeds the limit of proportionality
Precautions of safety	▪ Take care with masses and think the hanging masses would fall if the spring snapped. ▪ Wear eye protection
Precautions to improve accuracy or to give more reliable results	▪ Look perpendicular to the scale of the ruler while taking the readings to avoid parallax error ▪ Wait until the spring stops vibrating ▪ Always measure from the same point of spring ▪ Ensure the ruler is vertical/use horizontal aid ▪ Bench surface must be horizontal ▪ At least 5 loads for each spring if producing a graph ▪ Improve the accuracy by:

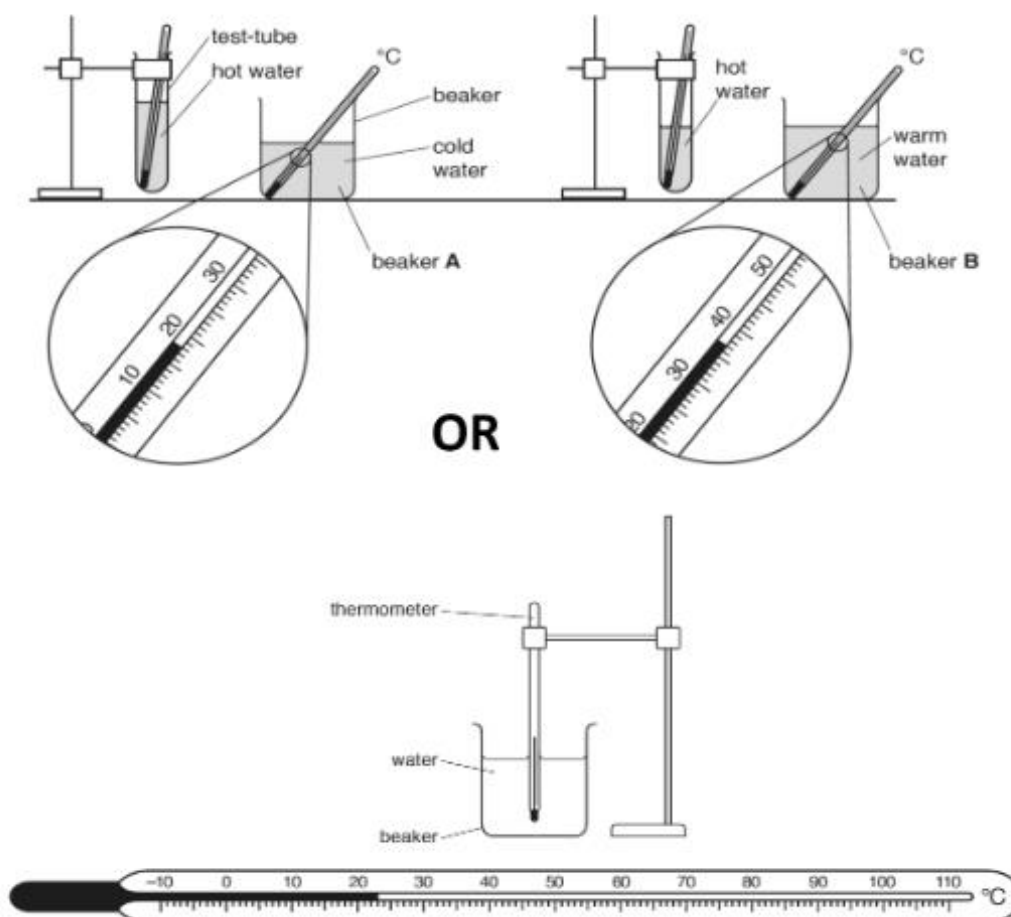
	➤ Clamping the metre ruler in place and then using the set square to make the length/extension measurement  ➤ Use the set square to make sure that the clamped ruler is vertical in relation to the bench ➤ Set the clamped ruler at 0 cm when no masses are added and ,so read the extension directly.
Conditions kept constant to fair comparison between springs of different materials	▪ Same spring diameter ▪ Same spring thickness ▪ Same range of loads ▪ Same length of spring/hanger ▪ Same number of coils ▪ Same coil spacing

1.4 Density (Displacement Method)



Aim	To measure the density of a solid
Theory	By definition, the density of a substance is given by: $\text{Density} = \text{mass}/\text{volume}$ For a regular-shaped solid the volume can be found by measuring its dimensions and the mass is determined by weighing. For an irregularly-shaped solid, the volume is found by immersion in a liquid.
Apparatus	As shown in figure.
Procedure	• Measure the mass of the solid material m on a top-pan balance • Fill the measuring cylinder with a known volume of water V_1 • Shack the solid material very well. • Immerse the solid in the cylinder. • Measure the new volume V_2 • Calculate the solid volume as: • $V = V_2 - V_1$ • Calculate the density as: $\text{Density} = m/V$
Precautions to improve accuracy or to give more reliable results	▪ Shack the solid material/rock ▪ Thin string/wire should be used ▪ Look perpendicular to the scale of the cylinder ▪ Look from the meniscus ▪ Put the solid material/rock gently
Possible causes of errors/inaccuracy	▪ Parallax error ▪ Student did not look from the meniscus of the liquid ▪ Splashes during immersing the solid material/rock ▪ Air bubbles may found in the solid material/rock ▪ String/wire used may be thick ▪ Cylinder may not be sensitive

2.1 Cooling rate experiment:

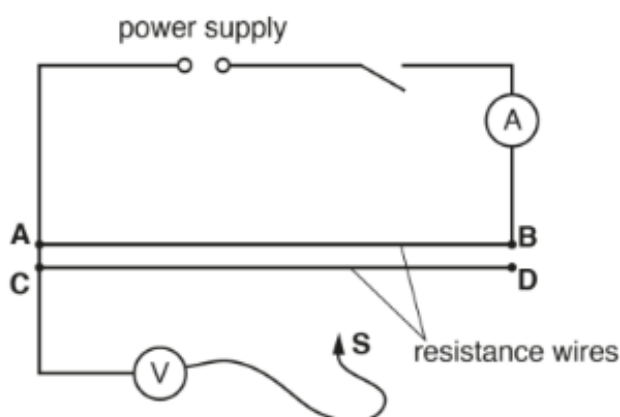


Aim	To study how the surrounding temperature affects the rate at which water/liquid cools
Theory	▪ A hot body loses its heat energy gradually to the surroundings. The experiment is done by measuring the temperature (using a thermometer) at equal time intervals (using a stopwatch). ▪ The temperature decreases until the body reaches the room temperature. ▪ The rate of cooling (rate of heat lost by radiation) of a body depends on: ➤ The difference in temperature between the body and surroundings

	➤ The exposed surface area of the body. The greater the area the greater the chance of the body to emit heat. ➤ The nature of the radiating surface (dull black is a good radiator while shiny or white surface is a poor radiator).
Apparatus	-heater-thermometer (or 2)- beaker(or 2)- stopwatch.
Procedure	▪ <i>The details of procedure may differ from an experiment to another (this is the general procedure).</i> ▪ Place heater into the beaker and turn it on to raise the temperature of water to..... ▪ Stir the contents of the water and place thermometer into the beaker. ▪ Note the starting temperature and turn on the stopwatch. ▪ Take readings of the thermometer and stopwatch at regular intervals. ▪ Tabulate and plot a graph to conclude your experiment.
Precautions of safety	▪ Take care when dealing with beakers of hot water. ▪ Set the hot water in a safe position where they will not be accidentally knocked over. ▪ Handle with caution to avoid burns
Precautions to improve accuracy or to give more accurate/reliable results	▪ Stir before reading ▪ Keep the thermometer at same level ▪ Set eye to as perpendicular to the scale ▪ Wait until reading stops rising (at start) ▪ Position clock so the thermometer and clock can be seen easily.
Possible causes of errors/inaccuracy and how to overcome	▪ Heat losses due to surrounds ➤ Lag container with insulator ➤ Cover container with lid ➤ Initial temperature near to room temperature
Conditions kept constant to fair comparison or to repeat the experiment	▪ Measure water into test-tube / beaker <i>to ensure the same volume/amount being used each time as cooling rates are different at different volumes.</i> ▪ Use the <i>same starting temperatures</i> in tubes <i>because cooling rates are different at different temperatures</i> ▪ Ensure all water in tube below level of water in beaker <i>to ensure that all water in tube has the same surrounding temperature.</i> ▪ Use insulation / lid on beaker <i>to keep water in beaker at (more) constant temperature</i>
	▪ Same beaker. ▪ Same thermometer <i>*Explanation is written in italic bold (needed in some questions).</i>

Common questions	▪ Explain why the rate of cooling decrease by time/ Why the initial rate of cooling is greater than the final rate of cooling? Because the liquid temperature becomes near the room temperature Justification: ➤ If you have two tables in the question, and you need to justify the statement, calculate the difference in temperature in a given time interval in each table then compare between the results within the limit of experimental accuracy. ▪ Examples of insulators: ➤ Glass, Wool, Cotton, Plastic, Rubber.
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3.1 Electricity experiment:



Aim	To measure the resistance of a wire and/or to investigate its dependence on the length of the wire.
Theory	The resistance R of a wire is given by: $R = V/I$ Where V is the p.d. across the wire and I is the current flowing through it.

Apparatus	-Battery-switch-wire or more- Ammeter-Voltmeter- meter rule- mounting clamps- 0-25 Ω rheostat (depending on the experiment)
Procedure	
Precautions of safety	The wire may become hot when current flows in it
Precautions to improve accuracy or to give more reliable results	▪ To overcome the heat effect in the wire: ➤ Use a battery of lower electromotive force. ➤ Switch on and off between readings. ➤ Add a lamp to lower the current ➤ Sometimes (not in all times)-depending on the given diagram- you can increase resistance of resistor. ▪ To improve accuracy and give reliable results: ➤ Look perpendicular while taking the readings. ➤ Tap on both the ammeter and voltmeter to check that the pointer is free to move (avoid sticking). ➤ Take several readings and then take overall average. ➤ Always check the connections are clean. ➤ Check the ammeter/voltmeter for zero-error ➤ Initially choose the highest range for the ammeter/voltmeter, then reduce the range, so that the deflection is almost full scale.
Possible causes of errors/inaccuracy	▪ Heating effect of the current ▪ The battery is used up ▪ Bad connection of the sliding contact

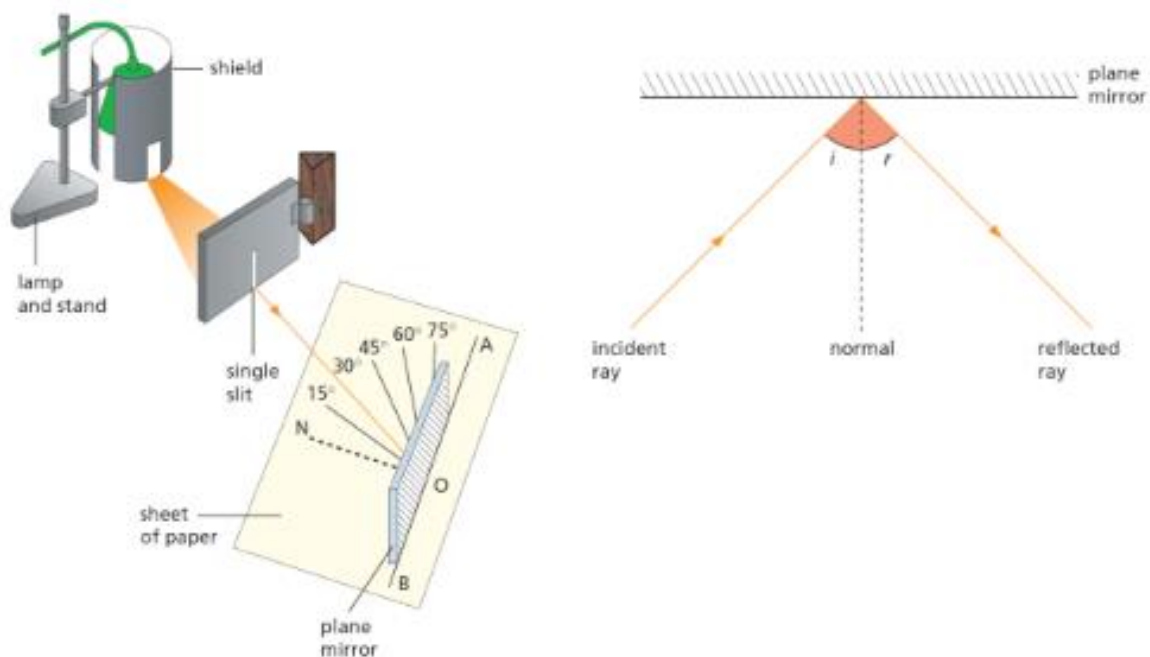
4.1 Optics experiment # 1 (focal length and linear magnification):



Aim	To determine the focal length and the linear magnification of a converging lens.
Theory	- Parallel rays from a distant object converge towards the principal focus of a converging lens. Note: the torch is not considered a distant object and is not used in this experiment while determining the focal length. The first part of this experiment should be performed without a torch, only depending on a distant object (may be outside the room/window) - The linear magnification, m, of an image is given by: $m = \text{height of image} / \text{height of object}$
Apparatus	Small torch, converging lens, screen/card, metre rule, adhesive/ mounting putty.
Procedure	- Focal length ➤ Step 1: Hold the lens up to a distant window and form a sharp image of the window on the screen/wall/a piece of card. ➤ Step 2: Measure and record the distance between the lens and card, the image distance, V. ➤ Repeat step 1 and 2 and obtain an average value for the image distance, V, which is equal to the focal length, f, of the lens. - Magnification:
Difficulties and how to overcome them	- Measuring the dimensions/height of the image using a hand ruler away from the image. - Can be overcome by using a translucent screen with a fixed ruler/grid.
Precautions to improve accuracy or to give more reliable results	- Dark room. - Object and lens have the same height from the bench. - Mark on lens holder to show position of lens centre. - Repeat and take the average. - Look perpendicular while taking readings. - Object/lens/screen perpendicular to the bench
Common questions	Draw/describe the image: ➤ Inverted (upside down) ➤ Magnification = $\frac{\text{distance of image from the length}}{\text{distance of object from the length}}$ (1) OR Magnification = $\frac{\text{Height of image}}{\text{Height of object}}$ (2) Use formula 1 or 2 depending on the question

	➤ Magnified if: Image distance > Object distance.  ➤ Smaller if: Object distance > Image distance. 
Common questions	▪ Why the object, the lens, and the screen should be perpendicular to the bench? To be able to achieve a sharp/complete/focused image.

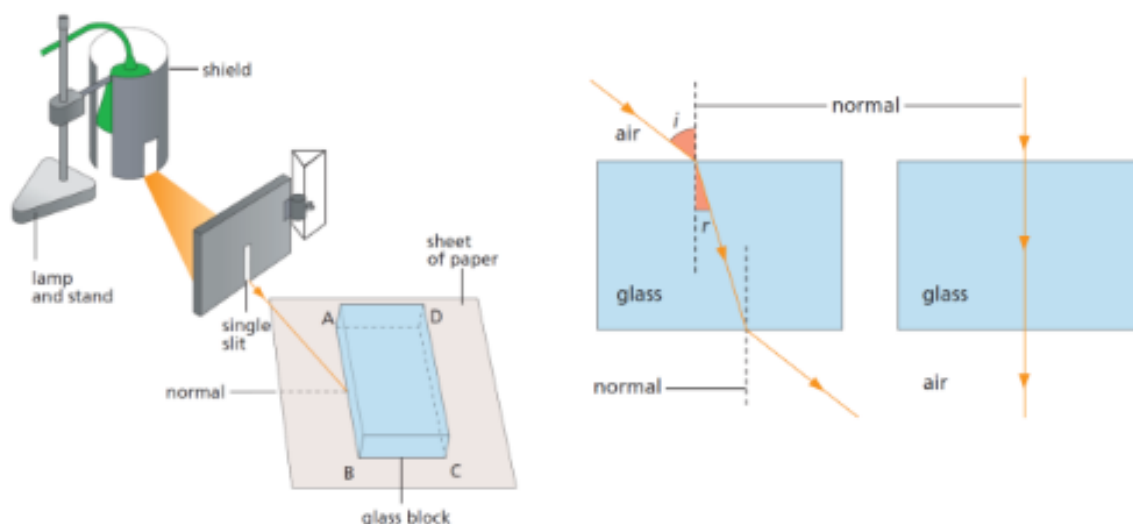
4.2 Optics experiment # 2 (Law of reflection):



Aim	To verify the law of reflection and identify the properties of an image in a plane mirror.
Theory	The law of reflection states that the angle of incidence equals the angle of reflection: $i = r$ Where i is the angle between the incident ray and the normal to the reflecting surface.
Apparatus	(Lamp + single slit) or a ray box, plane mirror, pins, protractor, adhesive

Procedure	- Place the plane mirror on a sheet of white paper, holding it vertical with plasticine or a block of wood. - Draw a line along the mirror. - Use the lamp and the single slit / ray box to shine a ray on the mirror. This is called the incident ray. You will see the reflected ray of light. Mark the position of each ray with optical pins. - Now take off all the apparatus and use a ruler to join up the optical pins, along the lines of rays of light. - Use a protractor or a set-square to draw a line at right angle (90°) to the mirror, as shown. This line is called the normal. - Use a protractor to measure the angles on each side of the normal.
Precautions to improve accuracy or to give more reliable results	Precaution while putting the pins: - Place the pins as far apart as possible (not less than 5 cm). - Use more pins. Place the pins vertical - Draw the lines so that they are as thin as possible - Look perpendicular while taking readings to avoid parallax error. - Use thin protractor - Look from the base of the pin: No concern about pins being vertical. - Base of the pin lies on the ray. - The base is always perpendicular to the plane - Repeat and average
Possible causes of errors/inaccuracy	- Thickness of lines - Thickness of protractor - Thickness of pins
	- Pin holes. - Thickness of mirror. - Glass on front of mirror causes double refraction.

4.3 Optics experiment # 3 (Refractive Index):



Aim	To determine the refractive index of glass and observe the path of light rays through a glass block.
Theory	The refractive index, n, of a medium is given by: $n = \frac{\sin i}{\sin r}$ Where i is the angle between the incident ray in air and the normal and r is the angle between the refracted ray and normal in the medium.
Apparatus	Lamp, single slit, glass block with lower surface painted white, protractor
Procedure	- Place a rectangular block of glass on a white sheet of paper. Trace the edge of the block using a pencil. - Draw a normal to one side of the block. - Let a light ray fall on the point where the normal is drawn. - Observe the emerging ray from the side of the block. - Put pins to determine the path of the incident ray and emergent ray. - Remove the block and draw lines representing the incident and the emerging rays.

	Measure the angle of incidence and angle of refraction of light then calculate the refractive index.
Precautions to improve accuracy or to give more reliable results	Precaution while putting the pins: - Place the pins as far apart as possible (not less than 5 cm). - Use more pins. Place the pins vertical - Draw the lines so that they are as thin as possible - Look perpendicular while taking readings to avoid parallax error. - Use thin protractor Look from the base of the pin: No concern about pins being vertical. - Base of the pin lies on the ray. - The base is always perpendicular to the plane - Repeat and average
Possible causes of errors/inaccuracy	- Thickness of lines - Thickness of protractor - Thickness of pins - Pin holes.

Exercises #1:

Some students are investigating the magnification produced by a converging lens. They are using the apparatus shown in Fig. 3.1.



Fig. 3.1

(a) A student sets the distance u between the illuminated triangle and the lens to 20.0 cm. She moves the screen until a sharp image of the triangle is seen on the screen. The student measures the height of the illuminated triangle h_o .

$$h_o = \dots\dots\dots 1.5 \text{ cm}$$

Measure and record, in Table 3.1, the height of the image of the triangle h_i on the screen, as shown in Fig. 3.2. [1]

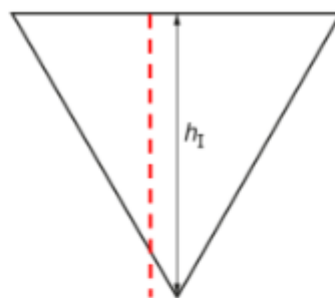


Fig. 3.2

Table 3.1

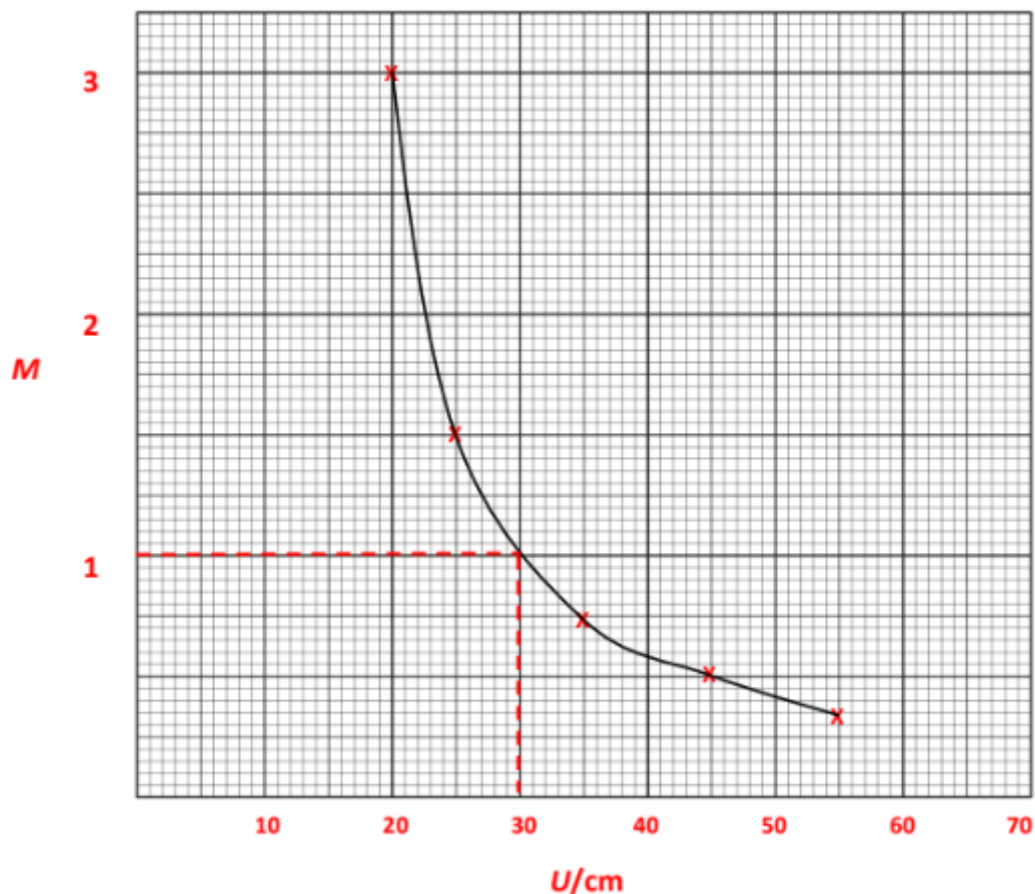
u/cm	h_i/cm	M
20.0	4.5	$4.5/1.5=3$
25.0	2.25	$2.25/1.5=1.5$
35.0	1.10	$1.10/1.5 = 0.73$
45.0	0.75	$0.75/1.5=0.50$
55.0	0.55	$0.55/1.5=0.37$

(b) The student measures the height h_i of the image for u values of 25.0 cm, 35.0 cm, 45.0 cm and 55.0 cm. Her results are shown in Table 3.1.

For each value of u , calculate and record in Table 3.1 a value for the magnification M . Use the equation $M = h_i/h_o$ and the value of h_o from (a).

[1]

(c) Plot a graph of M (y -axis) against u / cm (x -axis).



From your graph, determine the value of u when $M = 1.0$. Show clearly on your graph how you obtained the information.

$u =$ **30.0 cm**[2]

(e) Describe **one** difficulty that might be experienced when measuring the height of the image in this experiment. Suggest **one** improvement to the apparatus to overcome this.

difficulty **Measuring the dimensions/height of the image using a hand ruler away from the image**.....

improvement ... **Can be overcome by using a translucent screen with a fixed ruler/grid**..... [2]

(f) When setting up the apparatus, the student makes sure that the card with the illuminated triangle, the lens and the screen are all perpendicular to the bench. Explain why this is an important precaution in this experiment.

To be able to achieve a sharp/complete/focused image.....[1]

[Total: 11]

Exercise # 2:

Some students are investigating how the surrounding temperature affects the rate at which water cools.

They are using the apparatus shown in Fig. 1.1.

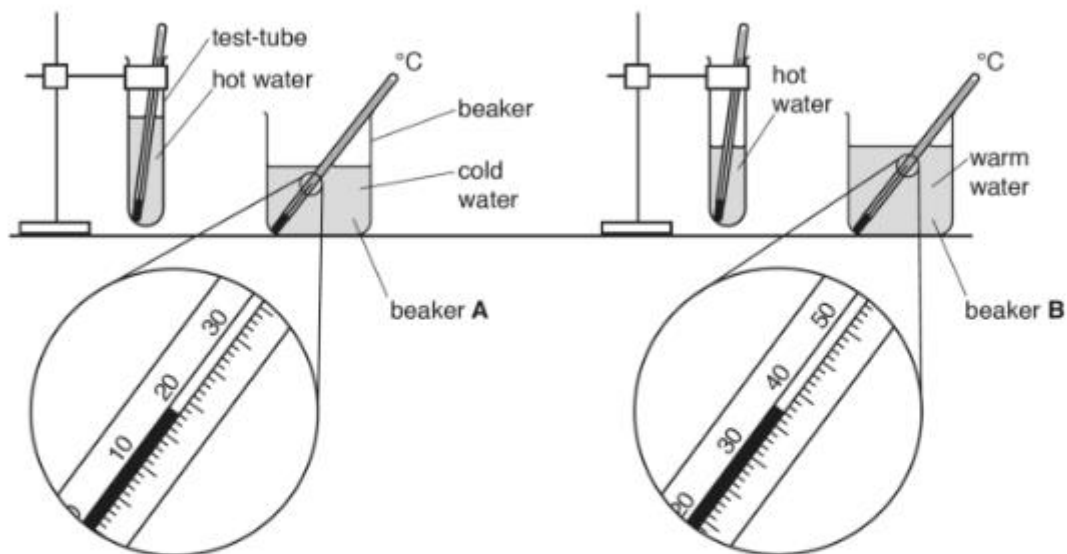


Fig. 1.1

(a) Using Fig. 1.1

- record the temperature θ_A of the cold water in beaker A,

$$\theta_A = \dots\dots\dots 18.0^\circ \text{C}$$

- record the temperature θ_B of the warm water in beaker B.

$$\theta_B = \dots\dots\dots 37.0^\circ \text{C}$$

[1]

- (b) The test-tubes of hot water are placed into beakers **A** and **B**.

The students record the temperatures θ of the water in the test-tubes every 30s. Their readings are shown in Table 1.1.

Complete the units and the time column in Table 1.1.

Table 1.1

time	tube in beaker A with cold water	tube in beaker B with warm water
$t /$	$\theta /$	$\theta /$
0	80.5	81.0
30	52.5	64.5
60	42.0	55.0
90	36.0	50.5
120	32.5	48.0
150	30.5	46.5
180	29.0	45.5

[2]

- (c) Describe **two** precautions that you would take, before reading the thermometer, to ensure that the temperature readings are as accurate as possible in the experiment.

1. **Stir before reading**.....

Set the eye perpendicular to scale.....

2. **Keep thermometer at same level**.....

Wait until reading stops rising at start.....

Position clock so that the thermometer and clock can be seen easily. [2]

- (d) Write a conclusion stating how increasing the temperature of the surrounding water affects the rate of cooling of the water in the test-tube.

Justify your answer by reference to the results in Table 1.1.

In the cold water beaker the change $\Delta\theta = 80.5 - 29.0 = 51.0^\circ\text{C}$ but in the warm water beaker the change $\Delta\theta = 81.5 - 45.5 = 35.0^\circ\text{C}$ therefore, the cooling rate increases when the surrounding temperature is lower and decrease when the surrounding temperature is higher.....

.....[2]

- (e) Suggest **one** change to the experiment shown in Fig. 1.1 to ensure that the comparison of the effect of surrounding temperature on cooling is a fair test.

Explain why the change is an improvement.

change **(1) Measure water in test tube / beaker**

..... **(2) Use the same starting temperature**

explanation **(1) To ensure the same volume/amount being used each time as cooling rates are different at different volumes**

..... **(2) Cooling rates are different at different temperature.**

[2]

- (f) The students use a measuring cylinder to measure 200 cm^3 of cold water.

Describe briefly how to read a measuring cylinder to obtain an accurate value for the volume of water. You may draw a diagram.

- **Reading must be taken perpendicular to scale**
- **Reading must be taken at the bottom of the meniscus.**

Exercise # 3:

A student is investigating how the material of a spring affects its behaviour when stretched.

The following apparatus is available to the student:

wires of different thickness, length and material
a set of 10g masses and a set of 100g masses, both with hangers
a wooden rod approximately 1 cm in diameter
other standard laboratory equipment.

Plan an experiment which will enable you to test the extension of springs made from different types of wire.

In your plan, you should include:

- instructions for making a spring from the wire that is provided,
- what you will measure,
- instructions for carrying out the experiment,
- the variables you will keep the same to ensure the comparison is a fair test,
- any precaution which should be taken or difficulty which might occur,
- how you will present your results.

You may draw a diagram if it helps to explain your plan.

- The spring can be made by winding a wire around the rod
- The extension of the spring will be measured against mass / load
- Repeat for springs of different materials / thickness
- Record the result in suitable annotated table and draw a graph
- Same spring diameter
- Same spring thickness
- Same range of loads
- Same length of spring / hangers
- Same coil spacing
- Difficulties / Precautions
 - The clamp retort stand might topple (use small loads)
 - Spring might overstretch / the spring is too weak (use loads that don't overstretch the spring)
 - Look perpendicular to the scale of the ruler while taking the reading to avoid parallax error
 - Wait until the spring stops vibrating
 - Make as small loop by the end of spring to apply load
 - Repeat each reading and take the average
 - At least 5 loads for each sample if producing a graph

Exercise # 4:

A student is investigating the resistance of three wires **A**, **B** and **C**. He is using the circuit shown in Fig. 2.1.

The circuit is set up to test wire **A**. The length, l of each wire is measured and recorded.

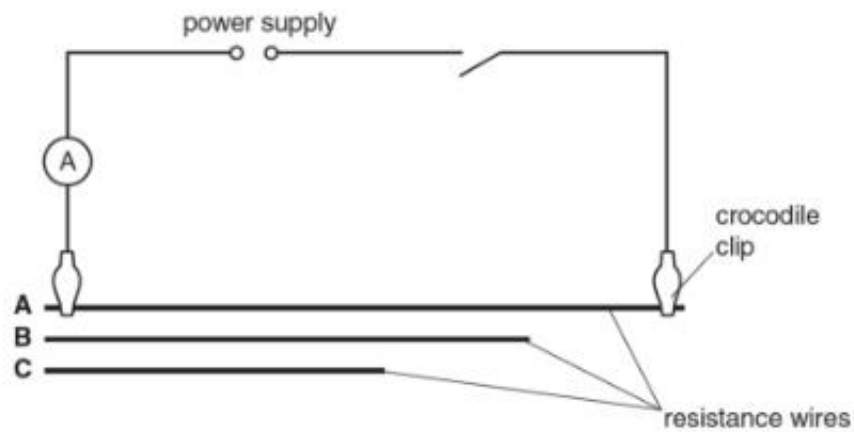
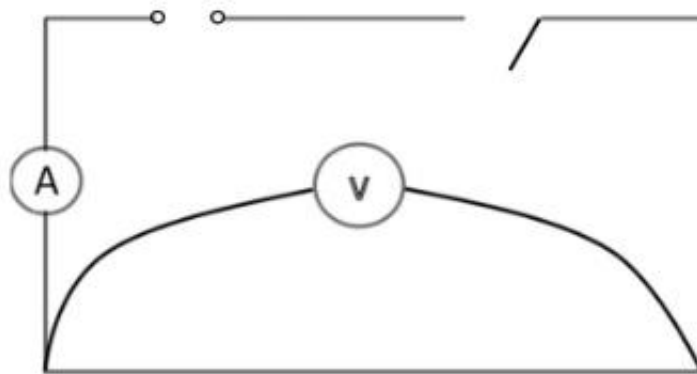


Fig. 2.1

- (a) On Fig. 2.1, draw a voltmeter connected so that it will measure the potential difference across wire **A**. [1]



- (b) In the first line of Table 2.1, record the potential difference V and current I for wire **A**, as shown in Figs. 2.2 and 2.3. [2]

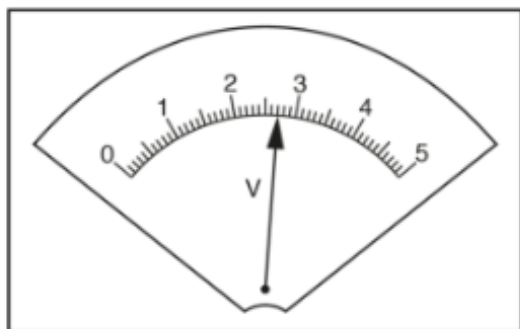


Fig. 2.2

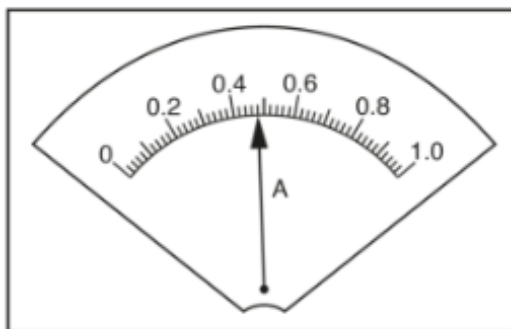


Fig. 2.3

Table 2.1

wire	l/m	V/V	I/A	R/Ω
A	0.900	2.7	0.48	5.63
B	0.500	2.4	0.75	3.20
C	0.400	2.2	0.85	2.59

- (c) The student connects the crocodile clips to wire **B** and then wire **C** in turn. His readings of potential difference and current are shown in Table 2.1.

Calculate, and record in Table 2.1, the resistance R of each wire.

Use the equation $R = \frac{V}{I}$.

$$R_A = 2.7/0.48 = 5.63 \, \Omega$$

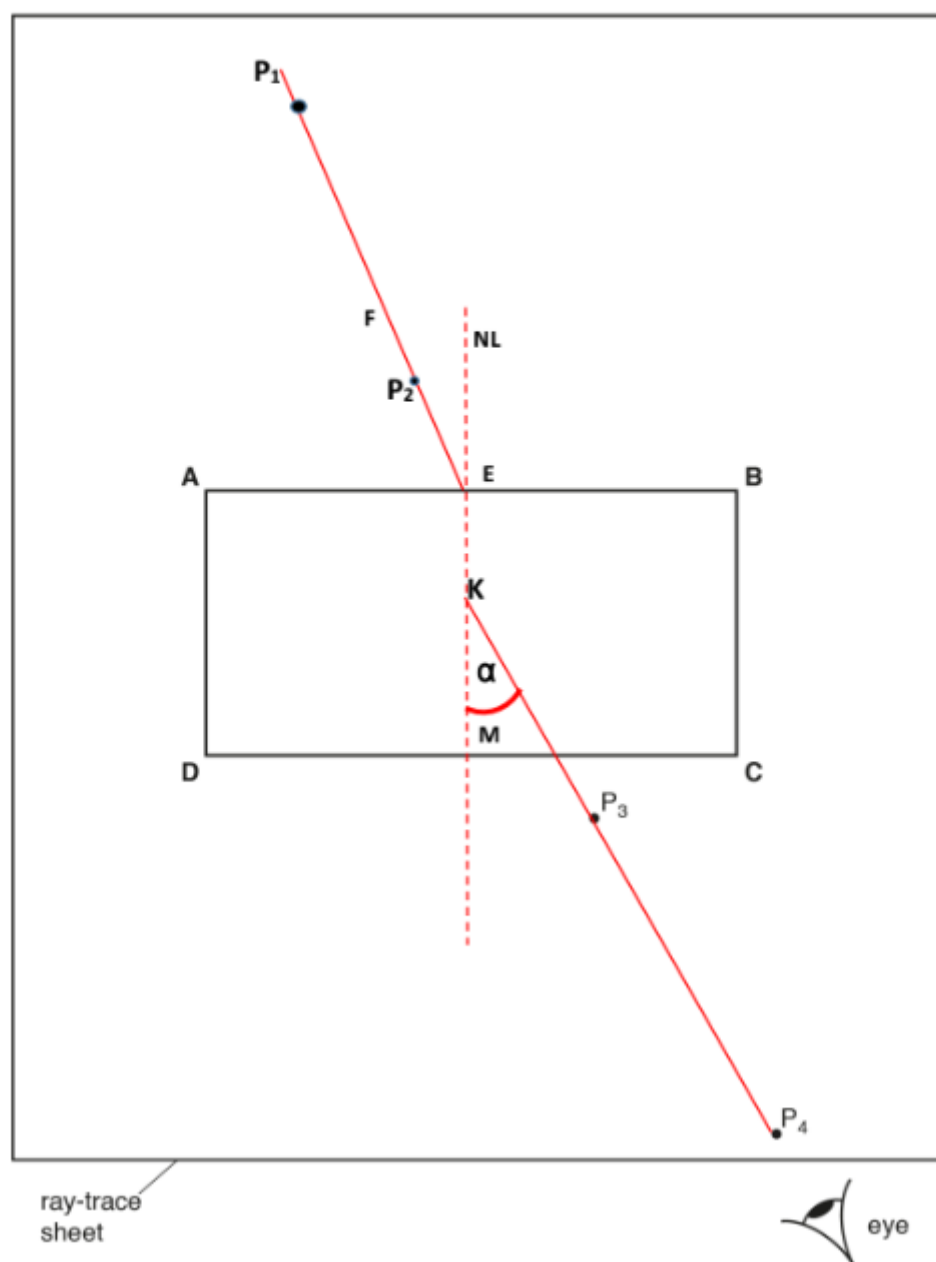
$$R_B = 2.4/0.75 = 3.20 \, \Omega$$

$$R_C = 2.2/0.85 = 2.59 \, \Omega$$

Exercise # 5:

The class is investigating the refraction of light passing through a transparent block. A student is using optics pins to trace the paths of rays of light.

Fig. 2.1 shows the student's ray-trace sheet.



- (a) • On Fig. 2.1, draw a normal at the centre of side **AB**. Label this line **NL**. Label the point **E** where the normal crosses **AB**. Label the point **M** where the normal crosses **CD**.
- Draw a line above **AB** to the left of the normal and at an angle of incidence $i = 30^\circ$ to the normal. Label this line **FE**.
- Label the positions of two pins P_1 and P_2 placed a suitable distance apart on **FE** for accurate ray tracing. [2]

- (b) The student observes the images of P_1 and P_2 through side **CD** of the block so that the images of P_1 and P_2 appear one behind the other. He places two pins P_3 and P_4 between his eye and the block so that P_3 and P_4 , and the images of P_1 and P_2 seen through the block, appear one behind the other. The positions of P_3 and P_4 are marked on Fig. 2.1.

Draw a line joining the positions of P_3 and P_4 . Continue the line until it meets the normal. Label this point **K**. [1]

- (c) • Measure and record the angle α between the line joining the positions of P_3 and P_4 and the line **KM**.

$$\alpha = \dots 28^\circ - 32^\circ \dots$$

- Measure and record the length x between points **M** and **K**.

$$x = \dots 20-24 \text{ mm} \dots$$
 [2]

- (d) The student repeats the procedure with the angle of incidence $i = 50^\circ$. His readings for α and x are shown.

$$\alpha = \dots 52^\circ \dots$$

$$x = \dots 19 \text{ mm} \dots$$

A student suggests that the angle α should always be equal to the angle of incidence i .

State whether the results support this suggestion. Justify your answer by reference to the values of α for $i = 30^\circ$ and $i = 50^\circ$.

statement **YES, the two angles are very close and almost the same**

justification $\% = \frac{\text{greatest angle} - \text{smaller angle}}{\text{greatest angle}} \times 100 = \frac{52 - 50}{52} \times 100 = 3.8 \%$

AND $\% = \frac{30 - 28}{30} \times 100 = 6.6\%$ the two percentages are within the experimental accuracy [2]

- (e) Suggest **one** precaution that you would take with this experiment to obtain reliable results.

- Place the pins as far apart as possible.
- Use more pins.
- Draw the lines so that they are as thin as possible
- Look perpendicular while taking readings to avoid parallax error. ...[1]
- Ensure pins are vertical/upright/erect

The class is investigating the motion of a pendulum.

Fig. 4.1 shows the apparatus.

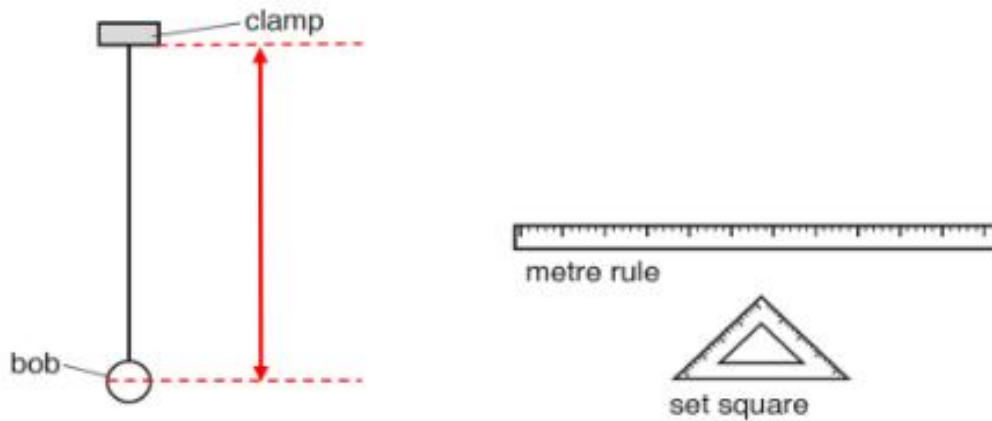


Fig. 4.1

- (a) (i) On Fig. 4.1, show clearly the length l of the pendulum. [1]
- (ii) Use Fig. 4.2 to explain how you would measure the length l accurately. You may draw on the diagram.

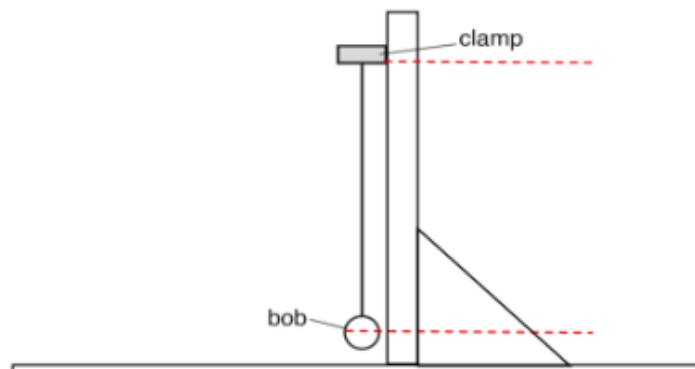


Fig. 4.2

- Put the ruler as close as possible to the pendulum
- Use the set-square as a reference to assure a vertical ruler
- The measurement should be from the bottom of the clamp

[2]

- (b) A student determines the period T of the pendulum. The period is the time taken for one complete oscillation. The student measures the time t for 20 oscillations.

Fig. 4.3 shows the time t .



Fig. 4.3

- (i) Calculate the period T of the pendulum.

$$T = \frac{20.22}{20} = 1.01 \text{ sec}$$

$$T = \underline{\underline{1.01 \text{ sec}}} \quad [1]$$

- (ii) Explain how measuring the time for 20 oscillations rather than one oscillation helps the student to obtain a more reliable value for the period.

To reduce the effect of start, stop error

Taking the average of 20 oscillations reduces the error in calculating one oscillation period and spreads the error over 20 swings

Time of one swing is too small to be measured [2]

- (c) The student wants to determine a value for the acceleration of free fall from his results. He needs the value of T^2 to do this.

Calculate T^2 .

Give your answer to a suitable number of significant figures and include the unit.

$$T^2 = \underline{\underline{1.02 \text{ sec}^2}} \quad [2]$$

[Total: 8]

$$T^2 = 1.01^2 = 1.02 \text{ sec}^2$$

EXPERIMENT: 1

A student is investigating whether using a lid reduces the time taken to heat a beaker of water to boiling point.

The student has the following apparatus available:

thermometer

250 cm³ glass beaker

250 cm³ measuring cylinder

heatproof mat

lid to fit the beaker

clamp, boss and stand.

Plan an experiment to investigate whether using a lid reduces the heating time.

You should

- list the additional apparatus that you would require
- explain briefly how you would carry out the investigation
- state the key variables that you would control
- draw a table, with column headings, to show how you would display your readings; you are not required to enter any readings in the table
- explain how you would use your readings to reach a conclusion.

A diagram is not required but you may draw a diagram if it helps your explanation.

ANSWER

clock / stopwatch

source of heat to boiling with and without lid measure time taken to reach boiling point

boil same volume / mass / amount of water same starting temperature

suitable table with column headings and units (seconds or minutes) conclusion drawn

EXPERIMENT: 2

A student is investigating how the material of a spring affects its behaviour when stretched.

The following apparatus is available to the student:

wires of different thickness, length and material

a set of 10 g masses and a set of 100 g masses, both with hangers

a wooden rod approximately 1 cm in diameter

other standard laboratory equipment.

Plan an experiment which will enable you to test the extension of springs made from different types of wire.

In your plan, you should include:

- instructions for making a spring from the wire that is provided,
- what you will measure,
- instructions for carrying out the experiment,
- the variables you will keep the same to ensure the comparison is a fair test,
- any precaution which should be taken or difficulty which might occur,
- how you will present your results.

You may draw a diagram if it helps to explain your plan.

ANSWER

apparatus:

MP1 springs made by winding wire around rod (or similar)

method:

MP2 apply load, measure length / extension of spring

MP3 repeat for spring(s) of different material

MP4 record results in suitable annotated table / bar chart / graph

control variables:

MP5 mark gained for any two of:

unstretched length of spring,

diameter of wire,

coil spacing,

load / range of loads used

diameter of spring

MP6 precautions / difficulties / additional points:

MP7 any two from:

clamp retort stand / might topple,

use small loads / spring might overstretch/spring too weak/use loads which don't overstretch
spring to support

loads

need to apply force smoothly / slowly,

suggested range of loads,

workable arrangement for applying load to spring (e.g. small loop at end of spring)

trial experiment to find (range of) loads to use

how to determine extension of spring,

repeat each reading and take average,

at least 5 loads for each sample if producing graph

EXPERIMENT: 3

A student is investigating the effect of draughts (moving air) on the rate of cooling of hot water.

The following apparatus is available to the student:

an electric fan with four speed settings

a supply of hot water

thermometer

250 cm³ beaker

250 cm³ measuring cylinder

stopwatch

clamp, boss and stand.

Plan an experiment to investigate the effect of draughts on the rate of cooling of hot water.

You should:

- explain briefly how you would carry out the investigation
- state the key variables that you would control
- draw a table, or tables, with column headings, to show how you would display your readings (you are **not** required to enter any readings in the table)
- explain how you would use your readings to reach a conclusion.

You may draw a diagram if it helps your explanation.

ANSWER

method to include:

MP1 measurements of temperature of hot water over a period of time/measurement of temperature at start and end of a

specified cooling time /measurement of time for a specified temperature drop

MP2 repeat using variety of fan speeds (blowing air over water surface)

MP3 AND MP4

two from:

room temperature

initial/starting temperature of hot water

volume/mass/amount of (hot) water

distance of beaker to fan for each speed setting

time of cooling (for a fixed temperature drop)

temperature drop (for a fixed time)

same beaker size/material

MP5 table with columns for fan speed, time, and temperature with units in the table headings (not the body of the table) for

time and temperature, but fan speed units not required

MP6 compare readings to find out which fan speed

produces the greatest temperature drop / takes least time **or** plot a graph of temperature against time

MP7 in the same time / for same temperature drop

or steepest gradient gives the fastest rate of cooling

EXPERIMENT: 4

Plan an experiment to investigate how increasing the number of layers of insulation affects the rate of cooling of hot water in a beaker.

Write a plan for the experiment, including:

- the apparatus needed
- what you would measure
- the variables you would keep the same to ensure the comparison is a fair test
- instructions for carrying out the experiment
- how you would present your results
- how you would use your readings to reach a conclusion.

You may draw a diagram if it helps to explain your plan.

ANSWER

method to include:

MP1 apparatus beaker with insulation and thermometer and stopclock (or alternative) mentioned

MP2 method pour hot water into container
measure temperature of hot water over period of time

MP3 repeat for additional layers

MP4 results: suitable table / graph / cooling curve

MP5 control variables any pair from:

same initial temperature,

same volume of water,

same size/material/thickness of beaker,

same thickness of each layer,

MP6 AND MP7

additional points any 2 from:

how cooling rate calculated/how to compare cooling curves,

read thermometer perpendicularly,

thermometer at same depth (for repeat) thermometer not touching beaker,

stir before reading thermometer,

use of lid,

minimum of 5 different thicknesses of insulation,

repeat experiment with different sized beakers/different amount of water,

sensible amount of water (50 cm³ to 500 cm³)

EXPERIMENT: 5

QUESTION

A student is investigating the effect of insulation on the rate of cooling of hot water in a 250 cm³ container.

The student can choose from the following apparatus:

thermometer

250 cm³ glass beaker

250 cm³ plastic beaker

250 cm³ copper can

250 cm³ measuring cylinder

three different insulating materials

clamp, boss and stand

stopwatch.

Plan an experiment to investigate the effectiveness of the three insulating materials.

You should

- explain briefly how you would carry out the investigation,
- state the key variables that you would control,
- draw a table, or tables, with column headings, to show how you would display your readings.

You are not required to enter any readings in the table,

- explain how you would use your readings to reach a conclusion.

A diagram is not required but you may draw a diagram if it helps your explanation.

ANSWER

MP1 Uses same container throughout

MP2 Hot water in container (any) and takes temperatures at intervals or at start and after a fixed time

OR Hot water in container (any) and takes time for a fixed temperature fall.

MP3 Repeats with different insulators (all three used)

MP 4&5 Any two from:

Constant room temperature

Same starting temperatures (clearly stated)

Same volumes of hot water (clearly stated)

Same thickness/amount of insulator

Use container without insulation

Use of a lid

Insulates bottom of container

Uses the copper can only

MP6 Table or tables as appropriate to method: Temperatures with unit °C and time with unit s (or min) and different insulators shown

MP7 Use of readings: graph of temperature against time

EXPERIMENT: 6

A student is investigating how the resistance of a wire depends on the length of the wire. The

student aims to plot a graph.

The following apparatus is available to the student:

ammeter

voltmeter

power supply

variable resistor

switch

connecting leads

resistance wires of different lengths

metre rule.

Plan an experiment to investigate how the resistance of a wire depends on the length of the wire.

You should

- draw a diagram of the circuit you could use to determine the resistance of each wire
- explain briefly how you would carry out the investigation
- suggest suitable lengths of wire
- state the key variables that you would control
- draw a table, or tables, with column headings to show how you would display your readings.

You are not required to enter any readings in the table.

ANSWER

MP1 Sample of wire must be clearly identifiable by a label on the diagram or by letters on the diagram with an explanation

in the text

MP2 All circuit symbols correct (even if circuit is incorrect)

Method:

MP3 Take readings of V and I

MP4 For 5 or more lengths

MP5 Range of lengths must be between 5 cm and 2 m with the largest length at least twice the smallest

Table drawn with headings:

MP6 l / m, V / V, I / A, R / Ω

Key variables to control:

MP7 Any one from

- Material / resistivity / conductivity / type of wire
- Diameter / radius / thickness / cross sectional area
- Temperature of wire

EXPERIMENT: 7

A student notices that the size of the image produced by a converging lens changes when the lens is moved further away from an object.

Plan an experiment to investigate how the size of the image varies with the object distance for a converging lens suitable for school experiments.

Write a plan for the experiment, including:

- a labelled diagram of the apparatus needed
- instructions for carrying out the experiment
- the factors that will limit the range of object distances
- the graph you will plot
- one precaution you will take to ensure reliable results, explaining what might be the effect of not taking this precaution.

ANSWER

apparatus:

diagram – lens, (illuminated) object, screen in suitable order for experiment
in line on flat surface

instructions:

set / measure object distance, move screen to get image, measure image height,
repeat for different object distances

limiting factor for range of object distances – one from:

- image virtual / too big for screen,
- image too dim / too small to measure,
- must be greater than focal length

graph:

image size / magnification against object distance

precaution:

any one suitable precaution and consequence of not taking it, e.g.

- dark room / bright light – image might not be distinct,
- lens and object at same height – image might not appear on screen,
- lens, object and screen perpendicular – image might be distorted,
- fix rule – may move and give incorrect distances
- mark position of lens on holder – cannot judge correct measurements / owtte
- detailed means of obtaining a sharp image – might not be correctly focused
- means of measuring image height accurately – might be obscured

EXPERIMENT: 8

A student is investigating resistors connected in parallel.

The following apparatus is available to the student:

ammeter

voltmeter
power supply
variable resistor
switch
connecting leads
a box of identical resistors.

Plan an experiment to investigate how the combined resistance of the resistors, connected in parallel, depends on the number of resistors.

You should:

- draw a diagram of the circuit you could use to determine the resistance of resistors connected in parallel (show only two resistors in your diagram),
 - explain briefly how you would carry out the investigation,
 - draw a table or tables, with column headings, to show how you would display your readings.
- You are not required to enter any readings into the table.

ANSWER

MP1 On circuit diagram: one voltmeter in parallel with any component

MP2 Circuit diagram correctly shows power supply, ammeter, unless in a branch, two or more resistors in parallel

MP3 Circuit diagram: Correct symbols for ammeter, voltmeter and fixed resistor

MP4 Repeat with a different number of resistors (in parallel)

MP5 Table that includes columns for number of resistors, voltage/V and current/A

MP6 & MP7 Then any two from:

Resistance calculated (may be shown in table)

Use low current (to stop resistors getting too hot)/switch off between readings

Use at least 5 different combinations

Repeat with different current or voltage or variable resistor setting

Drawing a graph of number of resistors against combined resistance

EXPERIMENT: 9

A student has noticed that different types of paper have different strengths.

Plan an experiment which will enable you to compare the strengths of different samples of thin Paper

Write a plan for the experiment, including:

- the additional apparatus needed
- instructions for carrying out the experiment, including any precautions you will take
- what you will measure
- how you will present your results
- how you will determine which paper is the strongest
- the variables you will keep the same to ensure the comparison is a fair test.

You may draw a diagram if it helps to explain your plan.

ANSWER

apparatus – workable arrangement

how applied force is measured

suitable table for results / plot a bar graph

how to conclude which is strongest

one suitable control variable:

e.g. same width of sample

same thickness / weight / length of paper

all samples fixed in same way

any 2 from:

control variable,

force applied smoothly / no jerking

ensure no tears before applying force

repeat for each type of sample / repeat with samples of different widths

soft mat under weights (to cushion fall) / clamp stand to bench

add weight of lower block to value of load

any other suitable precaution

EXPERIMENT: 10

A student suggests that the area of the water surface will affect the rate of cooling of hot water in a container.

Plan an experiment to investigate the relationship between surface area and rate of cooling.

Write a plan for the experiment, including:

- the apparatus needed
- how you will obtain a range of surface areas
- instructions for carrying out the experiment
- the measurements you will take
- the precautions you will take to ensure that the results are as reliable as possible
- the graph you will plot from your results – you should sketch the axes, with appropriate labels.

A diagram is not required but you may draw one if it helps to explain your plan

ANSWER

apparatus:

(set of) different sized beakers / containers, thermometer and stop clock / watch

method:

pour hot water into container (and allow to cool)

and measure temperature and time

repeat for a second container with a different surface area

precautions:

any two from:

same volume of hot water

same initial hot water temperature

same room temperature or other environmental condition

graph:

temperature change / rate of cooling against surface area,

temperature against time,

time to cool between fixed temperatures against surface area

additional point:

any one from:

- at least 5 different surface areas,
- sensible range of container sizes given,
- sensible amount of water stated,
- use of lagging / insulating material for container walls,
- same type of container
- how surface area may be calculated

EXPERIMENT: 11

A student is investigating whether the resistance of a wire depends on the material from which the wire is made. Resistance R is given by the equation $R = V/I$. The following apparatus is available to the student:

ammeter

voltmeter

micrometer screw gauge

power supply (0–3 V)

variable resistor

switch

connecting leads

wires of different materials.

Plan an experiment to investigate whether the resistance of a wire depends on the material from which it is made.

You should:

- draw a diagram of the circuit you would use to determine the resistance of each wire
- explain briefly how you would carry out the investigation, including the measurements you would take
- state the key variables that you would control
- draw a suitable table, with column headings, to show how you would display your readings (you are not required to enter any readings in the table).

ANSWER

MP1 Diagram showing power supply, ammeter, voltmeter and resistance wire correctly connected (variable resistor optional)

MP2 Correct symbols for ammeter and voltmeter. Variable resistor symbol correct if included.

MP3 Measure potential difference (voltage) and current and calculate resistance.

MP4 Repeat with other (types of) wires

MP5 Key variables length AND diameter stated

MP6 One of:

Repeat with different voltages (or currents).

Repeat and take average (voltage and current) readings.

Repeat entire experiment with different length or different diameter.

Use low current to prevent wire heating up.

Keep temperature of wire constant / switch off between readings

Use micrometer screw gauge to measure diameter / thickness of wire.

MP7 Table with columns for type of wire, voltage, current, resistance with correct units (V, A and Ω)

EXPERIMENTS

1) A student uses a pendulum to determine a value for the acceleration of free fall g .

(a) The student adjusts the pendulum until its length $l = 50.0$ cm. The length l is measured to the centre of the bob.

Explain briefly how the student avoids a parallax (line of sight) error when measuring length l .

- Either suitable use of a horizontal straight edge, explained briefly
- Or holding rule close to pendulum
- Or line of sight perpendicular (to rule)

(b) The student checks the value of the acceleration of free fall g in a text book. The value in the book is 9.8 m/s^2

(i) Suggest a practical reason experiment why the result obtained from the experiment may be different

- Explanation of cause of inaccuracy in measurement of t or l
- e.g. student did not react quickly enough when starting/stopping stopwatch OR difficulty
- measuring accurately to centre of bob

(ii) Suggest improvements to the experiment.

Any two from:

- Use different length(s)

- Repeat timing
- Use of a fiducial mark
- Increased number of oscillations
- Plot a graph using length and time or time²

(c) Explain why measuring the time for 20 oscillations, rather than for 1 oscillation, gives a more accurate value

- (reaction time) inaccuracy - smaller part of total time measured

2) A student is investigating the cooling of water.

a) Suggest conditions that should be kept constant for

- Room temperature (or other environmental condition)
- Temperature of cold water
- Temperature of hot water
- Volumes of water
- Size/shape/material/surface area of beaker

b) State precaution that you would take to ensure that the temperature readings are as reliable as possible

- Perpendicular viewing of scale OR stirring
- OR wait until temperature stops rising
- OR avoid delay (between adding water and taking temperature)
- Allow thermometer not touching beaker
-

c) Give reason why temperature of mixture is not equal to average temperature

- heat loss to surroundings/ beaker OR heat loss / drop in temperature by evaporation
- delays in taking readings
- reference to uncertainty in volume measurements

3) A student is determining the weight of a metre rule using a balancing method.

a) Suggest one practical reason why it is difficult to obtain exact results with this experiment.

- Difficulty of achieving balance or other sensible suggestion
- difficulty in achieving balance OR difficulty in positioning load exactly, e.g. load covers rule markings or uncertainty about position of centre of mass of load

b) This type of balancing experiment is difficult to carry out.

Suggest one practical difficulty and one way to try to overcome the difficulty. You may draw a diagram, if you wish.

- rule never quite balances,
- take average position / nearest to balance
- Rule unlikely to exactly balance/ difficulty for balance

- rule could slide on pivot
- mass could slide
- centre of mass of rule not at 50.0 cm mark
- rule not uniform

c) The calculation of W is based on the assumption that the centre of mass of the rule is at 50.0 cm mark.

Describe briefly how you would determine the position of the centre of mass of the rule.

- where rule balances on pivot

Describe how you would modify the experiment if the centre of mass was at the 49.7 cm mark

- Take readings from 49.7

4) A student is investigating the resistance of a lamp filament.

a) A student suggests that the resistance R of the lamp filament should be constant.

Suggest, referring to the observations, a reason why the resistance R may not be constant in this experiment.

- filament changes brightness, owtte
- increase / decrease / change in temperature of filament/lamp

5) A student determines the focal length of a lens.

a) Suggest precautions that you would take in this investigation in order to obtain reliable results.

- Use of darkened room / brighter lamp
- Mark position of centre of lens on holder
- Place metre rule on bench (or clamp in position)
- Ensure object and (centre of) lens are same height
- (from the bench)
- Object and lens and screen perpendicular to bench
- Move screen (slowly) back and forth to obtain best image (owtte)
- Ensure rule is touching object/lens/holder / screen or look perpendicular to ruler

b) Suggest two practical difficulties that may cause inaccuracy in the value of focal length obtained.

- difficulty deciding exact position of lens for best image / image not quite clear
- difficulty in measuring to centre of lens
- room too bright / lamp too dim

6) The class is investigating reflection using a plane mirror.

a) State two precautions that you would take in this experiment in order to obtain reliable readings.

- ensure pins are vertical / view bases of pins
- pins far apart (or > 5 cm)
- thin lines/sharp pencil / thin pins
- ensure mirror exactly on MR / ensure mirror does not move
- repeats

b) A student has done this experiment very carefully, taking these precautions.

She is disappointed to find that her line for the reflected ray is not exactly where she predicts from the theory.

Suggest a practical reason for this.

- thickness of lines / pencil/mirror/pins
- difficulty of lining up pins and images

7) The class is investigating the refraction of light passing through a transparent block.

a) Suggest one precaution that you would take with this experiment to obtain reliable results

- large pin separation
- ensure pins are vertical
- view bases of pins
- drawing thin lines / use a sharp pencil
- use thin pins

8) The student determines the average circumference of the cup, using a 50 cm length of string and a metre rule.

Shows how the student used the string to determine the average circumference.

a) Describe how you would use the string to obtain a more reliable value for the average circumference.

- measurement of circumference half way up, or at top and bottom
- more than one revolution used for the measurement in at least one position, and divide

9) A student carries out an experiment to compare how quickly thermal energy is conducted along rods made from different metals. Each rod is heated at one end with a Bunsen burner flame.

a) Suggest possible variables that the student should keep constant in order to make a fair comparison between the different metals.

- length of rod
- diameter/thickness / area (of cross-section) of rod
- amount of wax / type of wax
- weight/size/mass of marker
- position for the markers
- Bunsen) flame / (rate of) heating

- position of Bunsen / flame
- position of rod on tripod

10) The IGCSE class is investigating the cooling of a thermometer bulb under different conditions.

a) Suggest conditions that should be kept constant when this experiment is repeated.

- room temperature/ air conditioning/draughts / environmental conditions
- starting temperature (of thermometer) / temperature of (hot) water
- density of packing / amount of cotton wool / dryness of cotton wool

11) The IGCSE class is investigating the stretching of a spring.

a) When making measurements, the student is careful to avoid a line-of-sight error.

Suggest one other precaution that the student should take when measuring the distance

D between the bottom of the spring and the surface of the bench.

- Always measure from same point on spring (top or bottom of ring)
- Wait for spring/weight to stop bouncing
- Use of horizontal aid/ensure ruler is vertical
- Bench surface not uniform